Gamified interactive educational Learning Platform(Qubo)

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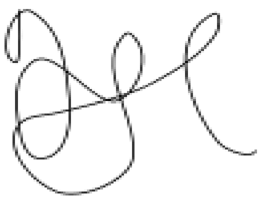
Kuala Lumpur

Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.



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Abstract

Malaysian SPM students often face challenges in identifying learning gaps and optimizing study strategies, leading to inefficient preparation and academic underperformance. This project proposes a Smart Learning Analytics Dashboard that monitors learning patterns and analyzes academic performance using machine learning techniques. The system incorporates spaced repetition algorithms, predictive analytics, and natural language processing to support automated quiz generation from textbook content.

The platform consists of several modules, including performance analytics, personalized study recommendations, content generation, and a lightweight gamification module designed primarily to help students relax and reduce study stress rather than serve as a core learning or practice mechanism. This relaxation-focused mini-game aims to maintain motivation and mental well-being while studying core SPM subjects, particularly those in the science stream.

The system is developed using an Agile software development methodology with iterative testing to ensure reliability, usability, and prediction accuracy. The expected outcome is an intelligent learning companion that assists students in improving engagement, identifying at-risk learners early, and providing data-driven insights to enhance study efficiency. Ultimately, the system aims to improve SPM examination performance and promote self-directed learning among Malaysian secondary school students.

Acknowledgement

First and foremost, we would like to express our deepest gratitude to our project supervisor, Mr. Ten Shai Cheong, for his invaluable guidance, unwavering support, and expert mentorship throughout this journey. His insightful feedback and constant encouragement enabled us to navigate challenges and deepen our understanding of the project's complexities. We are truly thankful for the time and dedication he invested in steering our work to completion.

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Lastly, I wish to acknowledge my partner—Yong Chao Juin. This project was a true collaborative effort, and our mutual dedication, shared responsibility, and constructive teamwork were fundamental to its successful outcome.

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Chapter 1

Introduction

1 Introduction

The Sijil Pelajaran Malaysia (SPM) examination represents a critical milestone in the Malaysian education system, serving as a gateway to tertiary education and career opportunities. However, students preparing for this high-stakes examination face numerous challenges that impede their learning effectiveness and academic success. Traditional learning approaches often fail to provide the personalized support and real-time feedback necessary for students to identify and address their knowledge gaps effectively.

In today's digital age, artificial intelligence and machine learning technologies offer unprecedented opportunities to transform education through personalization and data-driven insights. Qubo is designed to leverage these technologies to create an AI-powered personalized learning platform specifically tailored for SPM students. The platform aims to address the fundamental challenge of inefficient learning by providing intelligent analytics, personalized study recommendations, and automated content generation tools.

This project introduces a comprehensive learning analytics dashboard that employs spaced repetition algorithms, predictive analytics, and natural language processing to help students maximize their learning efficiency. By tracking performance patterns and analyzing learning behaviors, Qubo empowers students to take control of their academic journey and achieve better outcomes in their SPM examinations.

1.1 Objectives

1.1.1 To develop a centralized, interactive multimedia repository that curates 3D models and subject-specific video tutorials to enhance conceptual understanding

While AI and analytics can identify knowledge gaps, students require high-quality, accessible content to actually bridge those gaps. Qubo addresses this by implementing a structured multimedia library designed specifically for the SPM syllabus. This module moves away from static textbook learning by providing an interactive repository of Science-based 3D models and curated tutorial videos categorized by core subjects (Mathematics, Biology, Chemistry, Physics).

By integrating an embedded video player and an interactive 3D viewer directly into the platform, the system ensures that students can seamlessly consume visual and spatial educational content without leaving the learning environment. Furthermore, this repository allows students to curate their own personalized collections through "Favorites" and "Recent" tracking, empowering them to organize their study materials efficiently and revisit complex topics at their own pace.

1.1.2 To design and implement a lightweight gamification system that increases student engagement and facilitates the active application of scientific principles

Sustaining student motivation during rigorous SPM preparation is a persistent challenge. Traditional study methods can lead to cognitive fatigue and decreased retention. To counteract this, Qubo introduces a lightweight gamification module, specifically a turn-based Chemistry Card Game, designed to transform passive memorization into an active, engaging experience.

This system challenges students to apply their knowledge of the Periodic Table by combining elements to form valid chemical formulas in a simulated battle against a bot opponent. By categorizing chemical combinations into strategic actions (Attack, Defense, Heal), the game seamlessly integrates core chemistry rules with interactive gameplay mechanics. This objective ensures that students not only memorize scientific concepts but actively synthesize and apply them in a low-stakes, highly engaging environment, thereby reinforcing long-term retention.

1.1.3 To construct a secure administrative management dashboard that enables efficient resource curation and localized engagement monitoring

For a digital learning platform to remain relevant and accurate, it requires a robust backend infrastructure to manage its expanding repository of resources. Qubo incorporates a dedicated Resource and System Management module tailored for platform administrators. This dashboard provides a secure, centralized interface to curate and categorize external educational content, such as embedding YouTube tutorial links and uploading 3D assets, ensuring that the student-facing library remains high-quality and accurately tagged.

Additionally, this administrative system implements a localized analytics tracking mechanism to monitor internal video view counts, independent of external global metrics. This allows administrators to accurately quantify which resources are most effective and popular among Qubo users. Coupled with user moderation tools for account security and lockout management, this objective ensures the platform remains a secure, scalable, and data-informed environment for all users.

1.2 Background

Malaysian secondary school students face significant challenges in managing their learning effectively across multiple subjects as they prepare for the Sijil Pelajaran

Malaysia (SPM) examination. According to the 2024 SPM Results Analysis Report published by the Malaysian Examinations Board (Lembaga Peperiksaan Malaysia), while the overall pass rate remained stable at 97.5%, there are concerning trends in subject-specific performance that highlight the need for personalized learning interventions. For instance, Additional Mathematics recorded an alarming 24.9% failure rate with a GPMP of 5.62 despite having 19.9% of students achieving excellence grades. GPMP (Gred Purata Mata Pelajaran) means average subject grede point. Lower GPMP values (1.00 - 3.00) mean better overall performance (more students getting A+, A, A- grades) while higher GPMP values (4.00 - 7.00) mean weaker overall performance (more students getting lower grades like C, D, E, G). These statistics reveal a significant performance gap, suggesting that students struggle with identifying their knowledge gaps and allocating study time efficiently across different subjects.

The 2024 report further reveals critical disparities in academic achievement that underscore the inadequacy of one-size-fits-all teaching approaches. In core subjects like Bahasa Melayu and Bahasa Inggeris, the average grade points were 3.72 and 4.75 respectively, indicating room for improvement in language proficiency. More concerning is the performance in Science subjects: Physics recorded a failure rate of 0.4% with GPMP of 3.94, Chemistry showed 1.6% failure with GPMP of 4.52, and Biology had 0.8% failure with GPMP of 4.12. These varying performance levels across subjects demonstrate that students require differentiated support and personalized learning strategies to address their individual weaknesses effectively, particularly in subjects requiring higher-order analytical and problem-solving skills.

Furthermore, another research indicates that Malaysian students experience high academic stress, with 49.5% reporting moderate to high stress levels related to examinations (Ang & Huan, 2006). This stress is compounded by the traditional classroom approach that fails to accommodate diverse learning styles and paces (Don et al., 2015). Moreover, educators lack real-time visibility into individual student progress, limiting their ability to provide timely interventions. The 2024 analysis reveals that while urban schools generally perform better, rural students face additional challenges, with the urban-rural performance gap remaining significant across most subjects. For instance, in subjects like Ekonomi (GPMP 5.44) and Prinsip Perakaunan (GPMP 3.93), there is substantial variation in student performance, indicating that some students grasp concepts quickly while others struggle without adequate personalized support.

The COVID-19 pandemic has exacerbated these challenges, with the shift to online and hybrid learning exposing gaps in self-regulated learning skills (UNESCO, 2021). Current educational technology solutions in Malaysia are fragmented, focusing on either content delivery or assessment without integrating comprehensive analytics and personalized learning pathways (Ahmad et al., 2022). The 2024 SPM results demonstrate that students need more than just access to content, they require intelligent systems that can track their performance across multiple subjects, identify patterns in their learning behavior, and provide data-driven recommendations for improvement. With subjects ranging from core requirements to technical vocational courses showing varied performance outcomes, there is a clear need for an integrated platform that combines performance tracking, intelligent analytics, and personalized recommendations to support SPM students in their learning journey while providing educators with actionable insights for targeted interventions.

1.2.1 Comparision

Recent data from the Malaysian Ministry of Education reveals that only 31% of government schools have implemented any form of learning analytics system, and most of these are limited to basic attendance and grade tracking. Furthermore, SPM students have access to smartphones, only a few use educational apps regularly for exam preparation, citing lack of personalized content and engagement as primary barriers.

Table 1.1 below presents a comparative analysis of existing learning platforms available to Malaysian students, highlighting their strengths and limitations in addressing SPM preparation needs.

Table 1.1: Comparison of Existing Educational Platforms

Platform	Wayground (Old name: Quizizz)	Kahoot	Duolingo	Qubo (Proposed)
Analytics Dashboard	Basic	Limited	✔	✔
AI-Powered Quiz Generation	Pre-made only	Pre-made only	Adaptive	✔ Generated from user content
Predictive Analytics	✖	✖	✖	✔
Personalized Recommendations	Limited	✖	Algorithm-based	✔ Learning style-based
SPM-Specific Content	✖	✖	✖	✔
Gamification	✔	✔	✔	✔

This table 1.1 reveals that while existing platforms offer certain features, none provide the comprehensive, AI-driven, SPM-focused solution that Qubo delivers. The integration of advanced analytics, personalized recommendations, automated content generation, and predictive insights positions Qubo as a transformative tool for SPM preparation.

1.3 Target Market

1.3.1 Primary End User: SPM Students (Form 4 and Form 5)

The primary stakeholders are Malaysian secondary school students aged 16-17 preparing for their SPM examinations. With approximately 400,000 students taking the SPM annually, this represents a substantial user base with critical educational needs. These students will use the platform daily to complete AI-generated quizzes tailored to their current study materials, track their performance across all SPM subjects through comprehensive analytics dashboards, receive personalized study recommendations based on their learning patterns and performance data and access curated learning resources matched to their individual learning styles (visual, auditory, kinesthetic). Students benefit from enhanced self-awareness about their learning patterns, improved time management capabilities, and reduced examination anxiety through predictive insights and data-driven preparation strategies. The platform empowers them to take ownership of their learning journey and optimize their study efficiency.

1.3.2 Secondary End User: Educator and Teacher

Teachers and school administrators will utilize the platform to monitor class-wide performance trends and identify systemic knowledge gaps, identify at-risk students requiring immediate intervention through early warning alerts, access aggregated analytics to inform and refine teaching strategies, evaluate the effectiveness of different instructional approaches through performance metrics and provide targeted support and personalized interventions based on data-driven insights. Educators gain unprecedented visibility into learning outcomes beyond traditional assessments, enabling proactive rather than reactive teaching interventions. The platform reduces the administrative burden of manual progress tracking while providing sophisticated tools for differentiated instruction.

1.3.3 Secondary End User: Parents and Guardians

As supportive stakeholders, parents can monitor their child's study habits, consistency, and overall progress, understand specific subject areas requiring additional support or resources, receive timely alerts about performance concerns or significant improvements, support effective goal-setting and maintain productive study routines at home. Parents gain transparent access to their child's academic progress, enabling them to provide appropriate support, encouragement, and resources to facilitate successful SPM preparation.

1.4 Problem Statement

1.4.1 Problems

1. The Limitation of Static Learning Materials

SPM students heavily rely on traditional textbooks and scattered, unorganized online resources, which often fail to accommodate visual and spatial learners. There is a distinct lack of centralized, interactive educational platforms that seamlessly integrate multimedia resources—such as targeted video tutorials and manipulatable 3D models—preventing students from fully visualizing and comprehending complex scientific concepts.

2. High Cognitive Fatigue and Low Engagement

The rigorous and high-stakes nature of SPM preparation frequently leads to cognitive fatigue, stress, and a steep decline in student motivation. Traditional rote memorization techniques, particularly for intricate subjects like Chemistry, are often ineffective for long-term retention. Currently, there is an absence of accessible, lightweight gamified tools that allow students to actively apply scientific principles and practice their knowledge in a low-pressure, engaging environment.

3. Inefficient Content Curation and Inaccurate Engagement Tracking

As digital educational platforms scale, maintaining the quality and relevance of learning materials becomes a significant administrative burden. Without a dedicated, secure backend management system, educators struggle to efficiently curate external multimedia content, map them to correct syllabus subjects, and track localized student engagement. Relying on external global metrics (like public YouTube views) fails to provide accurate insights into which specific resources are actually benefiting the platform's user base.

1.4.2 Solutions

1. Interactive Multimedia Learning Repository and 3D Asset Integration

Qubo addresses the limitations of static textbooks by implementing a centralized, interactive multimedia library specifically curated for the SPM syllabus. This solution integrates an embedded video playback system and a manipulatable 3D model viewer directly into the platform, allowing students to visualize complex scientific and mathematical concepts without being redirected to external, potentially distracting websites.

By systematically categorizing these resources into core subjects such as Biology, Chemistry, Physics and enabling users to curate personal study collections via "Favorites" and "Recent" tracking, the system provides a highly organized, distraction-free environment. This ensures that students with visual and spatial learning preferences have immediate access to high-quality, syllabus-aligned content that directly supports the conceptual understanding required for SPM examinations.

2. Lightweight Gamification and Applied Science Mechanics

To combat study fatigue and enhance active recall, Qubo introduces a lightweight, interactive gamification module centered around a turn-based Chemistry Card Game. Rather than relying on the rote memorization of the Periodic Table, this solution requires students to strategically combine element cards to form valid chemical compounds in simulated battles against a bot opponent.

By translating correct chemical formulas into tangible gameplay mechanics categorized as Attack, Defense, and Heal actions, the system transforms abstract scientific rules into engaging, low-stakes challenges. This approach ensures that students do not merely memorize facts, but actively synthesize and apply their chemistry knowledge in real-time. This active application significantly improves long-term retention and sustains student motivation during rigorous exam preparation.

3. Centralized Resource Management and Localized Engagement Analytics

To ensure the platform's educational content remains accurate, scalable, and secure, Qubo incorporates a dedicated backend administrative dashboard. This centralized management system empowers platform administrators to seamlessly upload 3D assets, embed external tutorial links, and assign precise syllabus-aligned metadata such as subject tags without requiring direct database manipulation.

Furthermore, the solution features a localized analytics engine that tracks internal video engagement—bypassing skewed global metrics—to accurately identify which specific resources are truly benefiting the SPM student cohort. Coupled with robust user moderation tools and account security protocols such as managing lockout policies, this backend infrastructure guarantees that the student-facing modules are consistently fueled by a high-quality, data-driven, and secure learning ecosystem.

1.5 Advantages and Contributions

1.5.1 Advantages

1. Centralized Multimodal Learning Integration

Unlike generic video-sharing platforms or static educational websites, Qubo provides a highly focused, distraction-free multimedia repository. By natively embedding YouTube tutorials and interactive 3D science models into a single interface, the system eliminates the need for students to navigate external sites where they risk losing focus. This curated approach ensures that visual and spatial learners have immediate, organized access to SPM-aligned content, fundamentally improving how they visualize complex concepts.

2. Applied Interactive Gamification

While competitors like Quizizz and Kahoot focus on traditional, multiple-choice trivia, Qubo introduces a deeper level of applied gamification. The turn-based Chemistry Card Game requires students to actively synthesize their knowledge by forming valid chemical formulas to execute gameplay mechanics (Attack, Defense, Heal). This advantage shifts the learning paradigm from passive rote memorization to active, strategic application, significantly enhancing long-term retention of difficult scientific principles.

3. Localized Engagement Tracking and Secure Curation

A major advantage of Qubo's backend architecture is its decoupling of content management from global metrics. Instead of relying on skewed public data (like total YouTube views), the Admin Resource Manager tracks *localized* internal engagement. This provides educators and administrators with highly accurate, platform-specific data on which resources are actually benefiting the SPM student cohort. Combined with a secure interface for embedding links, uploading 3D assets, and managing user lockouts, the platform ensures a scalable, high-quality, and tightly controlled learning ecosystem.

4. Cultivation of Self-Directed Study Autonomy

Unlike rigid, linear e-learning platforms, Qubo's Personalized Library empowers students to take ownership of their revision process. By providing intuitive curation tools like "Favorites" collections and "Recent" history tracking alongside dynamic subject filters, the platform encourages students to build their own customized revision repositories. This fosters crucial self-directed learning habits, allowing students to efficiently revisit complex 3D models or tutorial videos exactly when they need them, rather than relying on a predetermined, inflexible curriculum pace.

5. Risk-Free Environment for Scientific Experimentation

The Chemistry Card Game provides a unique psychological advantage by significantly lowering the stakes of academic failure. In traditional SPM preparation, making a mistake on a practice paper can induce anxiety and lower self-confidence. However, testing an incorrect chemical combination against the game's bot opponent simply results in a strategic disadvantage within the match. This encourages a "trial-and-error" learning methodology, prompting students to experiment with chemical formulas and Periodic Table elements in a stress-free, engaging environment that naturally builds academic confidence.

6. Dynamic Scalability and Syllabus Adaptability

A common flaw in educational software is the rigid hardcoding of content, which causes the application to quickly become obsolete. Qubo's Resource & System Management module directly solves this by decoupling the educational content from the core application code. This provides the platform with immense scalability; as the Malaysian SPM syllabus evolves or new, higher-quality YouTube tutorials are published, administrators can instantly update the embedded links and 3D assets through the dashboard. This ensures the platform remains a continuously relevant, future-proofed educational tool without requiring ongoing developer intervention.

1.5.2 Contributions

To Students:

- **Enhances Conceptual Understanding:** Provides visual and spatial learners with direct access to manipulatable 3D models and curated videos, bridging the gap between abstract textbook theories and real-world visualization.

- **Reduces Study Fatigue:** Transforms the stressful, repetitive nature of SPM preparation into an engaging experience through interactive gameplay that rewards the practical application of knowledge.

- **Improves Focus:** Creates a "walled garden" of educational content through embedded video playback, protecting students from the algorithmic distractions of external platforms.

- **Promotes Organization:** Empowers self-directed learning by allowing students to build personalized study libraries through "Favorites" and "Recent" tracking features.

To Educators:

- **Streamlines Content Curation:** Dramatically reduces the administrative burden of managing digital resources by providing a simple, secure dashboard to tag, categorize, and upload multimedia content.

- **Delivers Actionable Insights:** Replaces guesswork with hard data by tracking internal resource views, allowing administrators to confidently identify and promote the most effective study materials.

- **Maintains Platform Integrity:** Equips administrators with direct user moderation tools, ensuring a secure environment by managing account lockouts and monitoring platform access.

To Community and Society:

- **Democratizes Access to Interactive Science:** Provides students from under-resourced schools with digital access to 3D scientific models and high-quality tutorials that would otherwise require expensive physical laboratory equipment.

- **Modernizes Malaysian Education:** Aligns with national goals to transition from traditional rote memorization toward interactive, technology-driven learning methodologies.

- **Fosters Sustainable Study Habits:** Reduces the high stress levels associated with SPM examinations by providing modern, engaging, and highly organized digital learning environments.

1.6 Project Plan

1.6.1 Project Scope

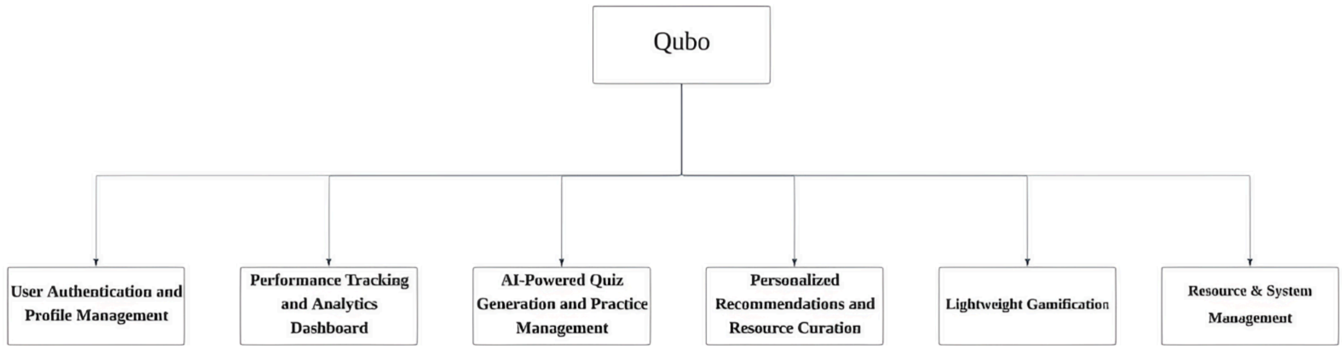

Figure 1.1: Modules of Qubo

The Qubo platform encompasses four primary modules, each designed to address specific aspects of SPM preparation and learning optimization:

Module 1: Personalized Recommendations and Resource Curation

The recommendation engine analyzes individual learning profiles to generate personalized curated learning resources. It categorizes students by learning style (visual, auditory, kinesthetic) and matches them with appropriate YouTube videos, articles, practice problems, and supplementary materials. The module provides daily study goals, topic prioritization recommendations, and optimal time allocation strategies. Gamification elements including streak tracking, achievement badges, and milestone celebrations maintain engagement and build consistent study habits.

Module 2: Lightweight Gamification

This module implements lightweight gamification through simple, beginner-friendly mini games. The purpose is to provide a relaxing activity after study sessions while increasing students’ interest, motivation, and confidence in the subject.

Module 3: Resource & System Management

This module serves as the centralized administrative backend for the Qubo platform. Its scope is strictly restricted to users with elevated Administrator privileges. It is designed to curate educational content, monitor platform-specific engagement, and enforce user security.

1.6.2 Milestone

Table 1.2: Project Milestones

ACTIVITIES	MILESTONE GOAL	START DATE	COMPLETION DATE
Approach and confirm project supervisor	Supervisor assigned and project scope confirmed	10/11/2025	19/11/2025
Prepare project proposal documentation	Complete proposal	20/11/2025	19/12/2025
Submit Individual Project Proposal	Approved proposal (Form 1, Form 2, Form 3)	10/11/2025	26/12/2025
Conduct literature review on AI/ML algorithms, learning analytics platforms, NLP for quiz generation, and EdTech solutions	Understanding of existing technologies and research gaps	27/12/2025	30/01/2026
Write Chapter 1: Introduction	Finalized introduction with problem statement, objectives, background, target market, and project scope	26/01/2026	06/02/2026
Research existing quiz generation systems and competitive landscape	Completed literature review covering AI/ML algorithms, learning analytics platforms, NLP for quiz generation, spaced repetition techniques, and existing EdTech solutions	09/02/2026	20/02/2026
Write Chapter 2: Research Background	Complete Chapter 2: Literature Review	09/02/2026	23/02/2026

Define system requirements and use cases	Defined Agile development process, technology stack (Python/TensorFlow, React, Supabase), hardware/software requirements, and detailed functional specifications for all modules	02/03/2026	05/03/2026
Develop project methodology	Clear development approach and analysis techniques defined	02/03/2026	10/03/2026
Write Chapter 3: Methodology and Requirements Analysis	Complete Chapter 3: Methodology and Requirements Analysis	02/03/2026	15/03/2026
Design system architecture and database schema	Finalized UI/UX wireframes, system architecture diagrams (API pipeline, database schema), API design specifications, and data flow documentation	16/03/2026	17/03/2026
Create UI/UX mockups and interface designs	Complete interface designs for all app screens	18/03/2026	27/03/2026
Write Chapter 4: System Design	Complete Chapter 4: System Design	16/03/2026	30/03/2026
Combine and format all chapters with cover page	Comprehensive compilation of Chapters 1-4 with proper formatting, references, and documentation	31/03/2026	13/04/2026
Setup development environment and project structure	Basic application framework ready	14/04/2026	20/04/2026
Develop quiz generation module	Functional quiz generation using AI/ML algorithms	21/04/2026	10/05/2026

Implement analytics dashboard	Working analytics dashboard displaying student performance metrics	11/05/2026	25/05/2026
Develop recommendation system	Recommendation system providing personalized learning paths	26/05/2026	05/06/2026
Integrate all modules into prototype	Functional prototype with core features (quiz generation, analytics dashboard) and documentation	06/06/2026	13/06/2026
Prepare test plan and test cases	Complete testing documentation and test cases for system validation	14/04/2026	18/06/2026
System preview with supervisor	Supervisor approval on system progress	14/04/2026	26/06/2026
Conduct final system testing	Fully integrated system with all modules (authentication, analytics, quiz generation, recommendations) verified and validated by supervisor	27/06/2026	10/07/2026
Submit draft FYP report	First draft documenting complete system implementation, testing results, analysis, and preliminary conclusions	11/07/2026	07/08/2026
Submission of Final FYP report and deliverables	Final report, complete source code, user documentation, deployment guide, and all digital deliverables submitted	08/08/2026	21/08/2026

1.6.3 Project Schedule Plan (Gantt Chart)

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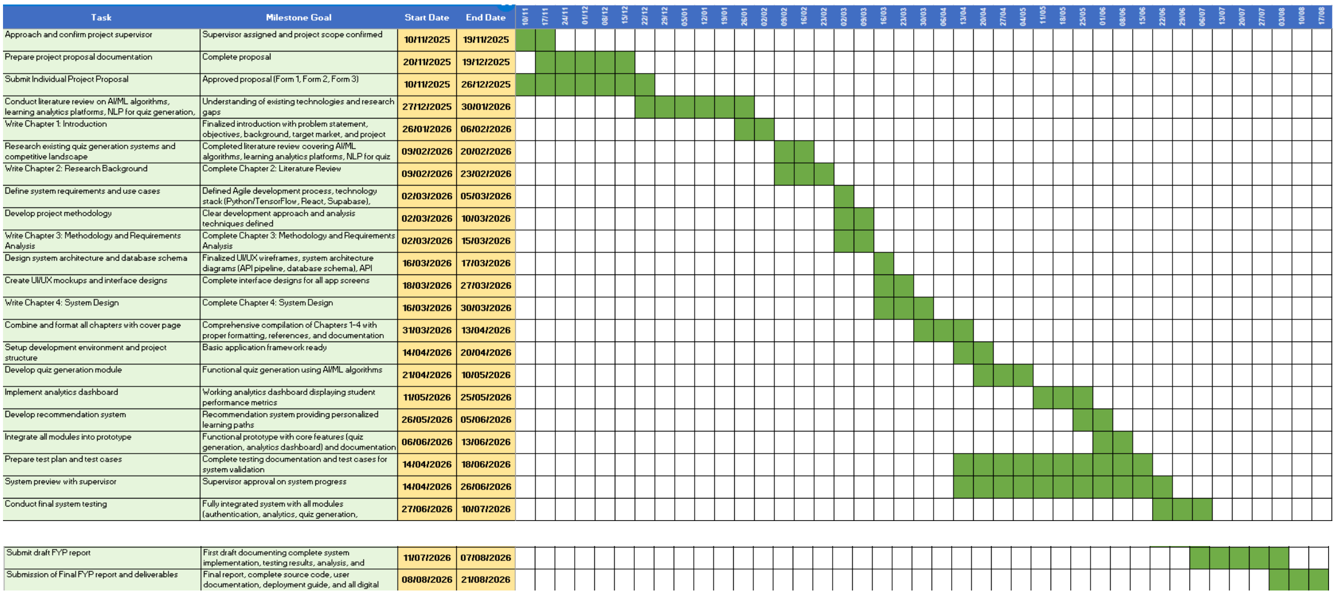

Task	Milestone Goal	Start Date	End Date
Approach and confirm project supervisor	Supervisor assigned and project scope confirmed	10/11/2025	19/11/2025
Prepare project proposal documentation	Complete proposal	20/11/2025	19/12/2025
Submit Individual Project Proposal	Approved proposal (Form 1, Form 2, Form 3)	10/11/2025	26/12/2025
Conduct literature review on AI/ML algorithms, learning analytics platforms, NLP for quiz generation,	Understanding of existing technologies and research gaps	27/12/2025	30/01/2026
Write Chapter 1: Introduction	Finalized introduction with problem statement, objectives, background, target market, and project	26/01/2026	06/02/2026
Research existing quiz generation systems and competitive landscape	Completed literature review covering AI/ML algorithms, learning analytics platforms, NLP for quiz	09/02/2026	20/02/2026
Write Chapter 2: Research Background	Complete Chapter 2: Literature Review	09/02/2026	23/02/2026
Define system requirements and use cases	Defined Agile development process, technology stack (Python/TensorFlow, React, Supabase),	02/03/2026	05/03/2026
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Write Chapter 3: Methodology and Requirements Analysis	Complete Chapter 3: Methodology and Requirements Analysis	02/03/2026	15/03/2026
Design system architecture and database schema	Finalized UI/UX wireframes, system architecture diagrams (API pipeline, database schema), API	16/03/2026	17/03/2026
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Prepare test plan and test cases	Complete testing documentation and test cases for system validation	14/04/2026	18/06/2026
System preview with supervisor	Supervisor approval on system progress	14/04/2026	26/06/2026
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Submit draft FYP report	First draft documenting complete system implementation, testing results, analysis, and	11/07/2026	07/08/2026
Submission of Final FYP report and deliverables	Final report, complete source code, user documentation, deployment guide, and all digital	08/08/2026	21/08/2026

Figure 1.2: Gantt Chart of Qubo

1.7 Chapter Summary and Evaluation

This chapter has provided a comprehensive introduction to the Qubo Gamified interactive educational Learning Platform, establishing the foundation for the project's development. The chapter began by contextualizing the challenges faced by Malaysian SPM students in an education system that increasingly demands interactive, engaging, and well-curated digital approaches to learning to supplement traditional study methods.

The three primary objectives of this specific system scope were articulated: developing a centralized interactive multimedia learning repository featuring manipulatable 3D models and curated videos, implementing a lightweight gamification system via an applied Chemistry Card Game, and constructing a secure administrative management dashboard for resource curation and localized analytics. Each objective addresses specific deficiencies in current educational technology offerings and directly responds to documented challenges in SPM preparation and platform scalability.

The background section established the educational context, highlighting research findings on student stress levels, cognitive fatigue, and the critical need for active learning environments. The comparative analysis of existing platforms demonstrated that while various solutions offer fragmented multimedia links or basic trivia games, none provides the cohesive, SPM-aligned, and actively gamified approach that Qubo delivers. The target market analysis identified primary users (Form 4 and Form 5 SPM students) and secondary stakeholders (platform administrators and educators), detailing how each group will benefit from the platform's specialized interfaces.

The problem statement section systematically documented three critical educational and administrative challenges: the severe limitations of static learning materials for visual and spatial learners, high cognitive fatigue coupled with low engagement during rigorous study sessions, and the administrative burden of inefficient content curation and inaccurate engagement tracking. For each problem, corresponding solutions were proposed, demonstrating how Qubo's integrated multimedia repository, applied gamification mechanics, and decoupled administrative backend directly address these technical and educational hurdles.

Finally, the advantages and contributions section highlighted Qubo's distinctive interactive features. Its centralized multimodal learning integration, applied interactive gamification, and localized engagement tracking demonstrate clear differentiation from competing platforms. The contributions to students, educators, and society were articulated, emphasizing the cultivation of self-directed study autonomy, the provision of a risk-free environment for scientific experimentation, and the platform's dynamic scalability to ensure a sustainable, long-term educational impact.

Finally, the project plan section defined the scope through four primary modules (authentication, analytics, quiz generation, and recommendations) and outlined eleven critical system functions. The milestone table provides a clear timeline for project execution, from proposal approval through final deliverable submission, ensuring accountability and systematic progress tracking.

This chapter successfully establishes the rationale, scope, and strategic direction for the Qubo project, providing a solid foundation for the detailed technical and methodological discussions that will follow in subsequent chapters. The comprehensive nature of this introduction ensures that all stakeholders (students, educators, supervisors, and evaluators) understand the project's significance, approach, and anticipated outcomes.

Chapter 2

Literature Review

2 Literature Review

2.1 Project Background

The Malaysian education system places significant emphasis on the Sijil Pelajaran Malaysia (SPM) examination, a high-stakes national assessment taken by Form 5 students that largely determines their academic and career trajectories. With over 400,000 candidates sitting for the SPM annually, the pressure surrounding this examination has given rise to widespread academic stress among secondary school students. A study by Ramli et al. (2018) published in the *Asian Journal of Psychiatry* found that approximately 49.5% of Malaysian secondary school students reported moderate to high levels of academic stress, with examination preparation cited as the primary stressor. This psychological burden is further compounded by students' limited ability to self-assess their own knowledge gaps, often resulting in inefficient and unstructured study habits.

Traditional classroom instruction in Malaysia operates predominantly within a one-size-fits-all framework, leaving little room for individualized learning. Hattie (2009), in his landmark meta-analysis *Visible Learning*, established that personalized, feedback-driven instruction produces among the highest effect sizes on student achievement, yet such approaches remain largely inaccessible in resource-constrained public school environments. This disparity is particularly acute in rural and semi-urban Malaysia, where students from lower-income households face additional barriers to quality supplementary educational resources. According to the World Bank, Malaysia's learning poverty rate stated that the proportion of children unable to read and understand a simple text by age 10 stood at approximately 48%, highlighting systemic gaps in foundational learning outcomes that persist into secondary education.

The rapid advancement of artificial intelligence and machine learning technologies has opened new possibilities for addressing these challenges through intelligent educational systems. Adaptive learning platforms powered by machine learning algorithms have demonstrated measurable improvements in student outcomes by dynamically tailoring content difficulty, pacing, and resource recommendations to individual learners. AI-driven personalization in education could improve learning efficiency by up to

20–40% compared to traditional instruction, underscoring the transformative potential of these technologies at scale.

Complementing adaptive learning, spaced repetition is a well-established cognitive science technique research on the forgetting curve, which demonstrates that information is retained more effectively when reviewed at systematically increasing intervals. Platforms such as Anki and Duolingo have successfully commercialized spaced repetition at scale; however, their content is largely generic and not aligned with the specific demands of the Malaysian SPM curriculum. This represents a critical gap in the current edtech landscape for Malaysian students.

Natural Language Processing (NLP) has similarly matured to a point where automated question generation from unstructured text is increasingly viable. Kurdi et al. (2020), in a systematic review of automatic question generation published in the *International Journal of Artificial Intelligence in Education*, found that NLP-based systems could generate educationally valid questions from textual sources with increasing accuracy, reducing the time educators spend on assessment creation and enabling scalable personalized practice. Despite this progress, no widely adopted platform currently enables Malaysian SPM students to generate curriculum-aligned practice questions directly from their own textbooks.

It is within this confluence of factors such as high examination pressure, inequitable access to personalized learning support, and the maturation of AI-driven educational technologies that Qubo is conceptualized. Qubo aims to bridge the gap between the advanced capabilities of modern edtech and the specific, underserved needs of Malaysian SPM students by integrating intelligent analytics, personalized recommendations, NLP-powered quiz generation, and lightweight gamification into a single, cohesive platform.

2.1.1 SPM Education Landscape and Student Challenges

Malaysian SPM examination system, enrollment statistics, subject requirements, and the documented academic pressures students face. Covers research on self-regulated learning deficiencies among Malaysian secondary students and how the lack of personalized feedback contributes to inefficient study habits and underperformance.

2.2 Literature Review

2.2.1 Cognitive Theory of Multimedia Learning

The integration of diverse media types in educational platforms is heavily supported by Richard Mayer's (2001) Cognitive Theory of Multimedia Learning (CTML). Mayer's research posits that the human brain processes audio and visual information through two distinct, parallel channels. His core finding, the Multimedia Principle, demonstrates that students achieve significantly deeper comprehension when learning from words and pictures combined rather than from words alone. Furthermore, the integration of interactive 3D manipulatives has been shown to significantly enhance spatial reasoning, which is particularly critical in STEM subjects where students must visualize microscopic or abstract structures (Dalgarno & Lee, 2010).

Limitation: A well-documented limitation in multimedia learning is "cognitive overload," which occurs when platforms present poorly integrated media or force students to navigate away from the primary learning environment to external sites (such as public video-sharing platforms). This causes a split-attention effect, increasing extraneous cognitive load and reducing overall learning efficiency (Sweller et al., 2011).

How Qubo addresses this: Qubo actively mitigates cognitive overload by natively embedding YouTube tutorial videos and manipulatable 3D Science models directly within a centralized, distraction-free repository. By keeping the student within the platform's ecosystem, Qubo ensures that visual and spatial information is processed seamlessly, maximizing the benefits of the Multimedia Principle while protecting the student from external algorithmic distractions.

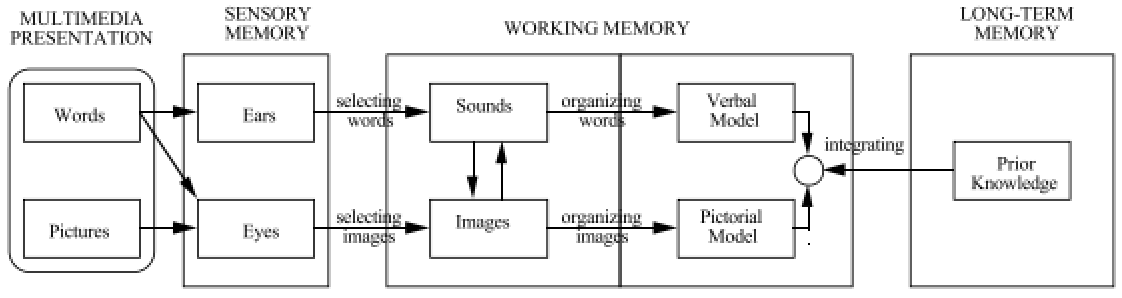


Fig. 8 Cognitive theory of multimedia learning (Mayer, 2001; Mayer et al., 2001)

2.2.2 Gamification Frameworks

Gamification is defined by Deterding et al. (2011) as "the use of game design elements in non-game contexts," and has been widely studied as a motivational strategy in educational technology. Common gamification elements include points, badges, leaderboards, progress bars, streaks, and achievement systems. Hamari et al. (2014) conducted a systematic review of 24 empirical gamification studies and found that while gamification generally produced positive motivational effects, outcomes were heavily dependent on the context of implementation and the target user group.

A prominent real-world example is **Duolingo**, which reported in 2020 that users who engaged with its streak and leaderboard features were 3.6 times more likely to complete daily learning goals compared to non-gamified users . Similarly, **Kahoot** reported that gamified quiz sessions increased student engagement and knowledge retention in classroom settings, with a 2019 study finding a 22% improvement in post-test scores when Kahoot was used versus traditional quizzes (Wang & Lieberoth, 2016).

However, research also highlights important caveats. Deci et al. (1999), in a meta-analysis of 128 studies on self-determination theory, found that **extrinsic rewards** such as points and badges can actually **undermine intrinsic motivation** over time if overused, a phenomenon known as the **overjustification effect**. This is particularly relevant for high-stakes examination contexts where sustained long-term motivation is critical.

Furthermore, research specifically addressing academic stress management in gamified environments is growing. Crocco et al. (2016) found that incorporating casual mini-games as deliberate relaxation breaks within study sessions reduced self-reported stress levels by 17% among secondary school students, while maintaining overall study session duration, suggesting that relaxation-oriented game elements serve a meaningfully different and complementary function to performance-oriented gamification.

How Qubo addresses this: Rather than deploying gamification as a core learning mechanic — which risks undermining intrinsic motivation through excessive extrinsic rewards — Qubo deliberately positions its mini-game module as a **stress relief and**

mental recovery tool. This design decision is directly supported by Crocco et al. (2016) and aligns with self-determination theory's emphasis on preserving intrinsic motivation as the primary driver of sustained learning engagement.

Based on the literature, Qubo incorporates a lightweight gamification module in the form of a relaxation-oriented interactive mini-game. The design of this module is informed by stress-relief gaming research rather than performance-based gamification, ensuring it complements rather than disrupts the core learning experience. The specific implementation approach for this module will be determined during the system design and development phase, subject to technical feasibility assessment."

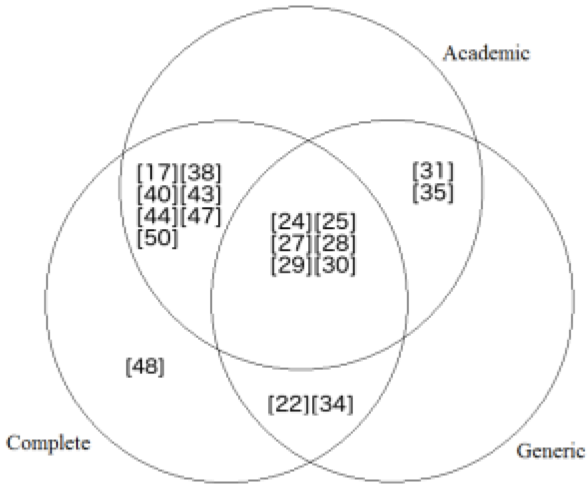


Fig. 2: Gamification framework's categorization

2.2.3 Recommendation Systems in Education

A recommendation system is an information filtering system that predicts and presents the most relevant content to individual users based on their preferences, behaviour, and historical interactions (Ricci et al., 2015). In educational contexts, Burke (2002) categorized recommendation approaches into three primary types: content-based filtering, which recommends resources similar to those previously engaged with; collaborative filtering, which identifies students with similar learning profiles and recommends resources that performed well for those peers; and hybrid filtering, which combines both approaches to overcome their individual limitations.

Content-based filtering performs well when sufficient metadata about learning resources is available, such as subject tags, difficulty levels, and resource types (Basilico & Hofmann, 2004). However, a well-documented limitation is the overspecialization problem, where the system only recommends resources similar to what the student has already consumed, restricting exposure to potentially more suitable learning approaches (Lops et al., 2011). Collaborative filtering addresses this by leveraging patterns across a broader student population. Klasnja-Milicevic et al. (2018) applied collaborative filtering to an e-learning platform and achieved a 31% improvement in resource relevance ratings compared to non-personalized content delivery, demonstrating the practical value of learner-to-learner pattern matching in academic contexts.

Learning style classification represents another critical dimension of personalized recommendations. The VARK model, developed by Fleming and Mills (1992), categorizes learners into Visual, Auditory, Reading/Writing, and Kinesthetic preferences, and remains one of the most widely operationalized frameworks in educational technology for learner profiling and resource matching (Rogowsky et al., 2015). Platforms such as Knewton have applied this model to route visual learners toward video content while directing reading-oriented learners toward structured text materials.

The integration of open web resources such as YouTube into recommendation engines has been explored as a cost-effective approach to expanding content libraries without proprietary content creation. Bulathwela et al. (2020) developed the TrueLearn model, which recommends open educational YouTube videos based on demonstrated learner knowledge states, achieving accuracy comparable to curated content platforms. This is particularly relevant for Malaysian SPM students from lower-income backgrounds who cannot afford commercial content subscriptions.

Regarding study scheduling, Kornell and Bjork (2007) demonstrated that students who followed algorithmically generated study schedules outperformed self-scheduled peers by an average of 74% on delayed retention tests, providing strong justification for incorporating time allocation recommendations as a core platform function rather than leaving study scheduling entirely to student discretion.

Limitation: A key challenge in recommendation systems is the **cold start problem**, where insufficient user data during early platform usage prevents the system from generating accurate personalized recommendations (Lika et al., 2014). Additionally, learning style classifications such as VARK have faced academic debate regarding their validity as fixed determinants of learning preference, with some researchers arguing that learning styles are dynamic rather than static traits (Rogowsky et al., 2015).

How Qubo addresses this: Similar to the ML module, Qubo mitigates the cold start problem by applying rule-based recommendations during initial usage, transitioning progressively to hybrid filtering as sufficient behavioural data is accumulated. VARK classification is treated as an initial preference indicator rather than a fixed label, allowing the system to dynamically adjust resource recommendations based on ongoing engagement patterns and performance data from the analytics module. Daily study goals and topic prioritization are driven by performance analytics, ensuring recommendations adapt as a student's knowledge state evolves. Streak tracking, achievement badges, and milestone celebrations are incorporated based on the findings of Hamari and Koivisto (2014) and Settles and Meeder (2016), with the intention of building consistent study habits rather than serving as a standalone gamification system.

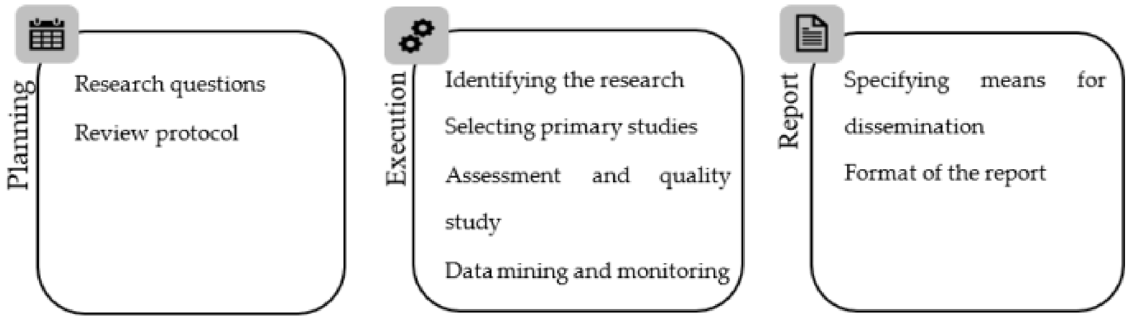


Figure 1. Phases of systematic review of literature.

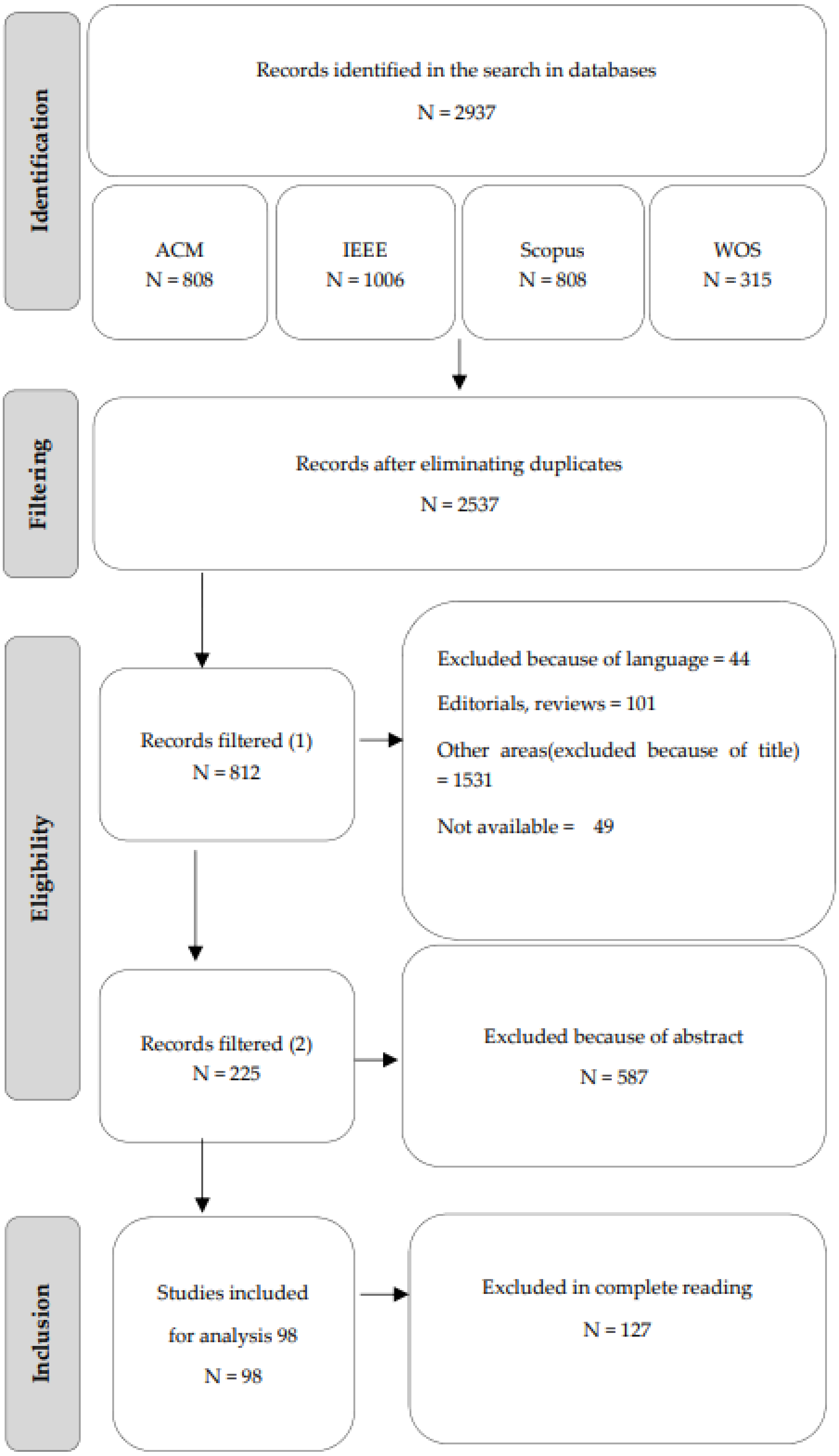


Figure 2. Flow chart showing study selection.

2.3 Feasibility Study

2.3.1 Technical Feasibility

State that Qubo will be developed using React (frontend), Python with TensorFlow (AI/ML backend), and Supabase (database). Justify each choice: React is widely adopted and well-documented, TensorFlow has proven ML model support, Supabase offers real-time data sync at low cost. Mention that NLP libraries such as HuggingFace Transformers are open-source and well-validated in research. Conclude that the technology stack is technically feasible.

2.3.2 Economic Feasibility

All core technologies (Python, TensorFlow, React, Supabase free tier, HuggingFace) are open-source or freemium. Development hardware is already owned. No licensing costs are required. Therefore the project is economically feasible with minimal overhead.

2.3.3 Operational Feasibility

Qubo only requires a smartphone or computer with internet access and a browser. Students can upload textbook images using their phone camera. The platform is designed to integrate naturally into existing study routines rather than replacing them. Therefore it is operationally feasible for the target user group.

2.4 Chapter Summary and Evaluation

This chapter provided a comprehensive review of the literature and foundational concepts underpinning the development of Qubo. The project background established the academic challenges faced by Malaysian SPM students, including high examination stress, limited access to personalized learning support, and the inability to accurately self-identify knowledge gaps. The system is conceptualized to bridge the gap between advanced educational technologies and these specific, underserved needs.The literature review examined the theoretical frameworks necessary to support Qubo's core interactive and analytical modules. It explored the Cognitive Theory of Multimedia Learning, demonstrating how the combination of visual and auditory information, alongside interactive 3D manipulatives, significantly enhances spatial reasoning and deep comprehension. Furthermore, the review established that embedding these resources natively mitigates the risk of cognitive overload caused by external algorithmic distractions. Gamification frameworks were also analyzed, revealing that while performance-based gamification can sometimes undermine intrinsic motivation, positioning interactive mini-games as deliberate relaxation tools successfully reduces academic stress. Finally, the review of recommendation systems validated the use of content-based, collaborative, and hybrid filtering techniques to effectively deliver personalized, syllabus-aligned resources.The feasibility study confirmed that Qubo is technically, economically, and operationally viable. By utilizing a modern stack comprising React, Python with TensorFlow, and Supabase, the system leverages open-source and freemium models that require no licensing costs, ensuring minimal economic overhead. Operationally, the platform integrates naturally into existing study routines, requiring only standard devices with internet access. These findings collectively establish the theoretical and practical foundation for the system design and development decisions presented in the subsequent chapters.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter presents the methodology and requirements analysis for the development of the Qubo platform. It begins by describing the Agile Software Development Methodology adopted in this project, followed by an explanation of the development process for the two assigned modules: the Personalized Recommendations and Resource Curation Module and the Lightweight Gamification Module. The chapter then outlines the functional and non-functional requirements that define the expected behaviors and quality attributes of the system. Finally, a summary is provided to consolidate the key points discussed throughout the chapter.

3.1 Methodology

3.1.1 Software Development Model

The development of Qubo follows the Agile Software Development Methodology. Agile is chosen because it supports iterative development, continuous testing, and incremental system improvement, which are suitable for developing interactive learning systems that involve recommendation features and gamified learning components.

Unlike traditional development models that follow a strict sequential process, Agile allows the system to be built in small functional increments, where each component can be designed, implemented, and tested before new functionality is added. This approach reduces development risk and allows improvements to be made based on testing feedback.

In this project, the development work is divided among team members. The scope of this study focuses on two main modules:

1. **Personalized Recommendations and Resource Curation Module**
2. **Lightweight Gamification Module**
3. **Resource & System Management**

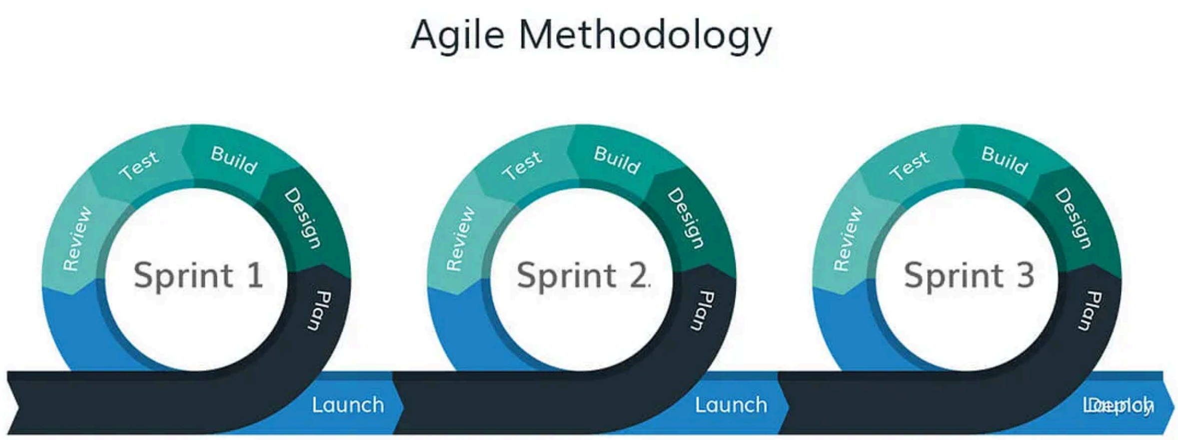
Each module is developed through four Agile stages: Requirements Analysis, System Design, Implementation, and Testing.

The Personalized Recommendations and Resource Curation Module is developed first because it provides the core learning functionality of the platform. This module organizes educational resources and suggests relevant learning materials based on user interaction.

After the recommendation module is implemented, the Lightweight Gamification Module is developed to enhance student engagement through an interactive chemistry-based game.

Once both modules are completed, they are integrated into the overall Qubo platform and tested to ensure that they function correctly with other system components developed by other team members.

Figure 3.1: Agile for Qubo Development



3.1.2 Personalized Recommendations and Resource Curation Module

The **Personalized Recommendations and Resource Curation Module** is responsible for organizing learning resources and providing users with relevant educational content. The objective of this module is to help students quickly locate suitable learning materials and encourage continuous learning through personalized suggestions.

The system collects basic interaction data such as **selected subjects, viewed videos, and favourite resources**. Based on these interactions, the system recommends relevant tutorial videos and learning materials to the user.

The recommendation mechanism uses a **content-based filtering approach**, where learning resources are categorized according to subject area and topic. When a user selects or interacts with a particular resource, the system identifies other resources that share similar categories and recommends them to the user.

This module also provides several resource management features, including:

- searching tutorial videos using keywords
- filtering resources by subject category
- saving favourite learning materials
- displaying recently viewed resources

These features allow users to manage their learning materials more efficiently and revisit previously studied content easily.

3.1.3 Learning Resource Processing

Before learning materials can be recommended to users, they must be organized and processed within the system. The Learning Resource Processing stage ensures that all tutorial videos and educational resources are structured in a way that supports efficient searching and recommendation.

Each learning resource stored in the system is associated with several metadata attributes, including:

- subject category (e.g., Chemistry, Physics, Mathematics, Biology)
- topic name
- resource type (video or interactive content)
- difficulty level

When a user interacts with a particular resource, the system records the activity in the database. The recommendation engine then analyzes this information to identify related learning materials with similar metadata attributes.

For example, if a user frequently watches videos related to chemical bonding, the system will prioritize recommending additional resources related to the same topic.

This processing approach allows the system to generate personalized learning suggestions while maintaining a simple and efficient recommendation mechanism.

3.1.4 Lightweight Gamification Module

The Lightweight Gamification Module introduces an interactive learning experience in the form of a Chemistry Card Battle Game. The purpose of this module is to increase student engagement and motivation by incorporating educational concepts into a game environment.

In the game, players use cards representing chemical elements from the periodic table. These elements can be combined to create different actions during gameplay.

Each combination produces a specific effect within the game system, such as:

- Attack actions that reduce the opponent's health
- Defense actions that protect the player from damage
- Healing actions that restore the player's health

The game follows a turn-based battle system, where players take turns selecting element cards and performing actions. The outcome of each move is determined based on predefined chemistry combination rules stored in the system.

The design of this module focuses on keeping the gameplay simple and educational, ensuring that students can understand the chemistry concepts while enjoying the game experience.

3.1.5 Resource & System Management Module

The Resource & System Management Module is developed to serve as the secure administrative backend of the Qubo platform. The objective of this module is to decouple content curation from the application's source code, allowing administrators to dynamically update educational materials without requiring developer intervention.

This module focuses on three core administrative functions: content management, engagement tracking, and user moderation. For content management, the system provides interfaces for administrators to input external YouTube URLs, which the system automatically formats for embedded <iframe> playback and upload 3D science assets. The system ensures all resources are accurately tagged with SPM subject metadata such as Biology and Chemistry to fuel the Personalized Recommendation module.

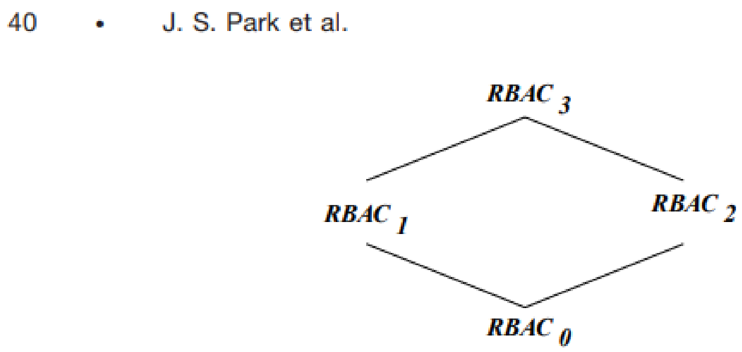
Furthermore, this module implements a localized analytics engine. Instead of relying on global metrics like public YouTube views, the system tracks internal student engagement, internal view counts to generate accurate, platform-specific trending data. Finally, a security dashboard is implemented to monitor user authentication and manage account lockouts triggered by the system's security protocols.

3.1.6 System Integration

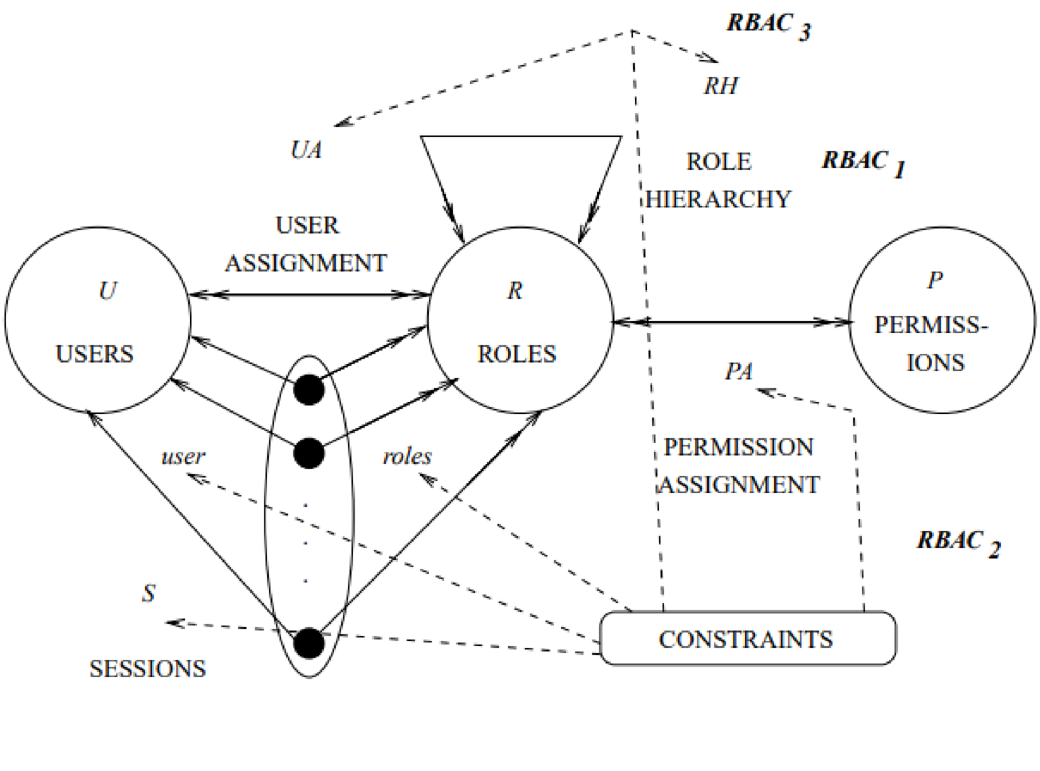
After all three modules are implemented, they are integrated into the overall Qubo platform. Integration ensures that the student-facing personalized learning features and the gamification module operate seamlessly alongside the secure administrative backend within the same system environment.

The frontend interface, developed using modern web technologies, allows students to access learning resources and the gamification module through a unified navigation system. Simultaneously, the administrative backend connects directly to the platform's centralized database. This ensures that any educational resources such as embedded YouTube links or 3D models uploaded and categorized by the Admin in Module 3 are immediately accessible and correctly filtered in the student's Personalized Library (Module 1).

This integration allows the system to accurately track localized user learning activities across different modules, securely storing engagement data to power both the student's personalized recommendations and the Admin's analytics dashboard. Testing is performed during the integration stage to verify that all three modules function correctly, data flows accurately between the backend and frontend, and role-based access control (RBAC) (Park, Sandhu, & Ahn, 2001) successfully prevents unauthorized access to the management dashboard.



(a) Relationship among RBAC models



(b) RBAC models

3.1.7 System Evaluation

The completed modules are evaluated to ensure that they meet the functional and non-functional requirements defined earlier in the project.

Three main aspects are evaluated during system testing:

1. **Recommendation Effectiveness**
   The recommendation module is evaluated based on its ability to provide relevant learning resources to users. This is measured by observing whether recommended resources are related to the user's selected subjects and previously viewed materials.

2. **Gamification Functionality**
   The gamification module is evaluated to ensure that the chemistry card battle game operates correctly. Testing verifies that element combinations trigger the correct actions, including attack, defense, and healing effects according to the established chemistry rules.

3. **Administrative Functionality and Analytics Accuracy**
   The Resource & System Management module is evaluated to ensure that administrators can successfully embed multimedia links, assign metadata, and manage user accounts. Testing strictly verifies the accuracy of the localized analytics engine—ensuring internal view counts increment correctly upon student interaction—and confirms that security lockouts operate as intended.

User and administrative testing is conducted to ensure that all modules provide a smooth experience for their respective target audiences and function as intended. Any issues discovered during testing are corrected before final system deployment.

3.2 Requirement Analysis

3.2.1 Functional Requirement

Module	Functional Requirement
1.0 Personalized Recommendations and Resource Curation	1.1 The system shall display 3D Model Page from library.
	1.2 The system shall display Tutorial Video Page From library.
	1.3 The system shall allow the user to go to his/her respective 3D Model page from library.
	1.4 The system shall allow the user to go to his/her respective Tutorial Video page from library.
	1.5 The system shall display the user Science base model based on what science subject user want from the 3D model page.
	1.6 The system shall display the category filter in what subject user choose.
	1.7 The system shall display the Tutorial Video homepage with a list of trending and popular videos upon entering the page.
	1.8 The system shall display a search bar that allows the user to search for tutorial videos by keyword or title.
	1.9 The system shall display subject filter categories (e.g. Mathematics, Biology, Chemistry, Physics) on the Tutorial Video page.
	1.10 The system shall filter and display tutorial videos based on the subject selected by the user.
	1.11 The system shall display a "You" section that contains the user's personal video collections.
	1.12 The system shall display a "Recent" list that shows tutorial videos recently viewed by the user.
	1.13 The system shall display a "Favourites" list that contains tutorial videos saved by the user as favourite.
	1.14 The system shall allow the user to add or remove a tutorial video from his/her Favourites list.
	1.15 The system shall allow the user to share a tutorial video to another user within the system.
	1.16 The system shall display the video title, subject tag, duration, and thumbnail for each tutorial video.
	1.17 The system shall allow the user to play a tutorial video directly within the application.

	1.18 The system shall recommend tutorial videos based on the user's learning profile and recently viewed history.
2.0 Lightweight Gamification	2.1 The system shall display the Chemistry Card Game page as the main gamification module after a study session.
	2.2 The system shall allow the user to start a new game session against a bot opponent.
	2.3 The system shall randomly deal a hand of Periodic Table element cards to the player at the start of each turn.
	2.4 The system shall display each element card with its symbol, name, and atomic number (e.g. H, Hydrogen, 1).
	2.5 The system shall allow the player to select and combine multiple element cards to form a valid chemical combination.
	2.6 The system shall categorize each valid chemical combination into one of three types: Attack, Defense, or Heal.
	2.7 The system shall display the combination type, effect value, and chemical formula when the player selects a valid combination.
	2.8 The system shall provide a Combination Guide that displays all available chemical combinations within the game for player reference.
	2.9 The system shall categorize the Combination Guide by Attack, Defense, and Heal sections.
	2.10 The system shall allow the player to access the Combination Guide at any time during the game session.
	2.11 The system shall implement a turn-based battle system where the player and bot take turns alternately.
	2.12 The system shall display both the player's and bot's HP bar at all times during the battle.
	2.13 The system shall reduce the bot's HP based on the Attack combination value played by the player.
	2.14 The system shall reduce the player's HP based on the bot's attack value during the bot's turn.
	2.15 The system shall restore the player's HP based on the Heal combination value played by the player.
	2.16 The system shall reduce incoming damage to the player based on the Defense combination value played by the player.
	2.17 The system shall declare the player as the winner when the bot's HP is reduced to zero (0).
	2.18 The system shall declare the bot as the winner when the player's HP is reduced to zero (0).
	2.19 The system shall simulate the bot's turn by randomly selecting a valid chemical combination from its dealt cards.
	2.20 The system shall display the bot's played combination and its effect to the player

	after each bot turn.
	2.21 The system shall display a game result screen showing Win or Lose at the end of each game session.
3.0 Resource & System Management	3.1 The system shall provide a secure login interface restricted to Administrator accounts only.
	3.2 The system shall allow the Admin to upload, edit, and delete YouTube tutorial video links.
	3.3 The system shall allow the Admin to assign metadata, including Video Title and Subject Tag, to tutorial videos.
	3.4 The system shall allow the Admin to upload, update, and categorize 3D Science models.
	3.5 The system shall automatically format Admin-provided YouTube URLs into embedded links for in-platform playback.
	3.6 The system shall track and record the number of times a tutorial video is played internally within the platform.
	3.7 The system shall display an analytics dashboard showing a ranked list of trending videos based strictly on internal view counts.
	3.8 The system shall allow the Admin to view a list of registered user accounts.
	3.9 The system shall alert the Admin to user accounts that have been temporarily locked due to exceeding three (3) failed login attempts.
	3.10 The system shall allow the Admin to manually reset and unlock restricted user accounts.

3.2.2 Non-Functional Requirements

Non-Functional Requirements	
1.0 Reliability	1.1 The system shall achieve a minimum **system uptime of 98%**, excluding scheduled maintenance periods.
	1.2 The system shall ensure that learning resources such as tutorial videos and 3D models remain accessible without interruption during normal usage.
2.0 Performance	2.1 The system shall load main interface pages within 5 seconds under normal network conditions.
	2.2 Tutorial videos shall begin playback within 3 seconds after user selection.
	2.3 The system shall load 3D model learning pages within 5 seconds.

	2.4 The Chemistry Card Game module shall process player actions and game responses within 2 seconds.
	2.5 Search results for tutorial videos shall be displayed within 3 seconds after query submission.
	2.6 The system shall load the Admin Resource Management analytics dashboard and aggregate data tables within 5 seconds under normal network conditions.
3.0 Accuracy	3.1 The system shall ensure that recommended tutorial videos are relevant to the user's selected subject category.
	3.2 The system shall ensure that chemical combinations in the card game follow correct chemistry rules based on the periodic table elements used in the game system.
	3.3 The system shall ensure that the localized engagement analytics (view counts) accurately isolate internal platform activity from global YouTube metrics without data synchronization errors.
4.0 Security	4.1 User authentication shall restrict system access to registered users only.
	4.2 The system shall enforce a maximum of three (3) failed login attempts before temporarily locking the account.
	4.3 User personal data and learning activity records shall be securely stored in the database.
	4.4 The system shall implement Role-Based Access Control (RBAC) to ensure that the Resource & System Management dashboard is strictly inaccessible to standard student accounts.
5.0 Usability	5.1 The system interface shall be simple and intuitive so that secondary school students can use it without external guidance.
	5.2 The system shall support multi-language interface options.
	5.3 The system shall display clear confirmation or feedback messages after every user action, such as saving favourites or starting a game session.

	Navigation between learning modules (videos, 3D models, and game) shall be accessible through a consistent menu structure.
	5.5 The administrative dashboard shall present data management tools and analytics in a structured, organized layout suitable for non-technical educators or administrative staff.

3.2.3 Development Environment

Table 3.3: Development Environment

Resources	Tools
Hardware	Laptop Specification: ● I5-13420H CPU ● NVIDIA GeForce GTX 4050 ● 16 GB RAM ● 2 TB SSD Peripheral ● Mouse ● Keyboard
Software	● Visual Studio Code
Programming Language	● React (frontend) ● Tailwind CSS (CSS framework) ● Python with TensorFlow (AI/ML backend) ● Unity (Game develop)
Database	● Supabase
API	● Gemini Flash API
Version Control	● Git / GitHub
Operating System	Windows 11

3.3 Chapter Summary and Evaluation

This chapter discussed the methodology and requirements analysis for the three assigned modules of the Qubo platform. The Agile Software Development Methodology was adopted as the development framework due to its iterative nature, which supports continuous testing and incremental improvement throughout the development process.

The methodology was applied across the development stages of the Personalized Recommendations and Resource Curation Module, the Lightweight Gamification Module, and the Resource & System Management Module, followed by system integration and evaluation.

The functional requirements defined for Module 1 established the core behaviors of the learning resource features, including the 3D Model library, subject filtering, and personal collection systems. The functional requirements for Module 2 defined the complete mechanics of the Chemistry Card Battle Game, covering chemical combination categorization and turn-based battle logic. Finally, the functional requirements for Module 3 established the secure administrative capabilities necessary to curate embedded multimedia content, track localized engagement metrics, and manage user account security. Together, these requirements provide a clear specification of what the system must perform to meet both the learning needs of SPM students and the operational needs of platform administrators.

The non-functional requirements addressed five quality dimensions: Reliability, Performance, Accuracy, Security, and Usability. These requirements ensure that the system remains stable during normal usage, responds within acceptable time limits, applies correct chemistry logic, protects user data, and delivers an intuitive interface. Overall, the requirements defined in this chapter are comprehensive, utilizing the Agile methodology to provide a solid foundation for the subsequent system design and implementation phases.

Chapter 4

System Design

4 System Design

This chapter outlines the comprehensive system design and architectural framework for the Qubo educational platform. It translates the functional and non-functional requirements established in Chapter 3 into concrete structural blueprints. The design illustrates a modern, web-based architecture utilizing a React frontend, a Python-based backend, and a Supabase database infrastructure. Subsequent sections detail the User Interface (UI) design to demonstrate the user experience, followed by a series of Unified Modeling Language (UML) diagrams—including Use Case, Sequence, Entity-Relationship (ERD), Class, and Deployment diagrams—to clearly model the system's behavioral logic, data storage, and physical deployment structure.

4.1 User Interface Design

https://www.figma.com/design/wwLIL2rJ82IkVQRwgwEvwj/FYP---System-design?m=auto&t=WRsQpcE4yoz81YwF-1

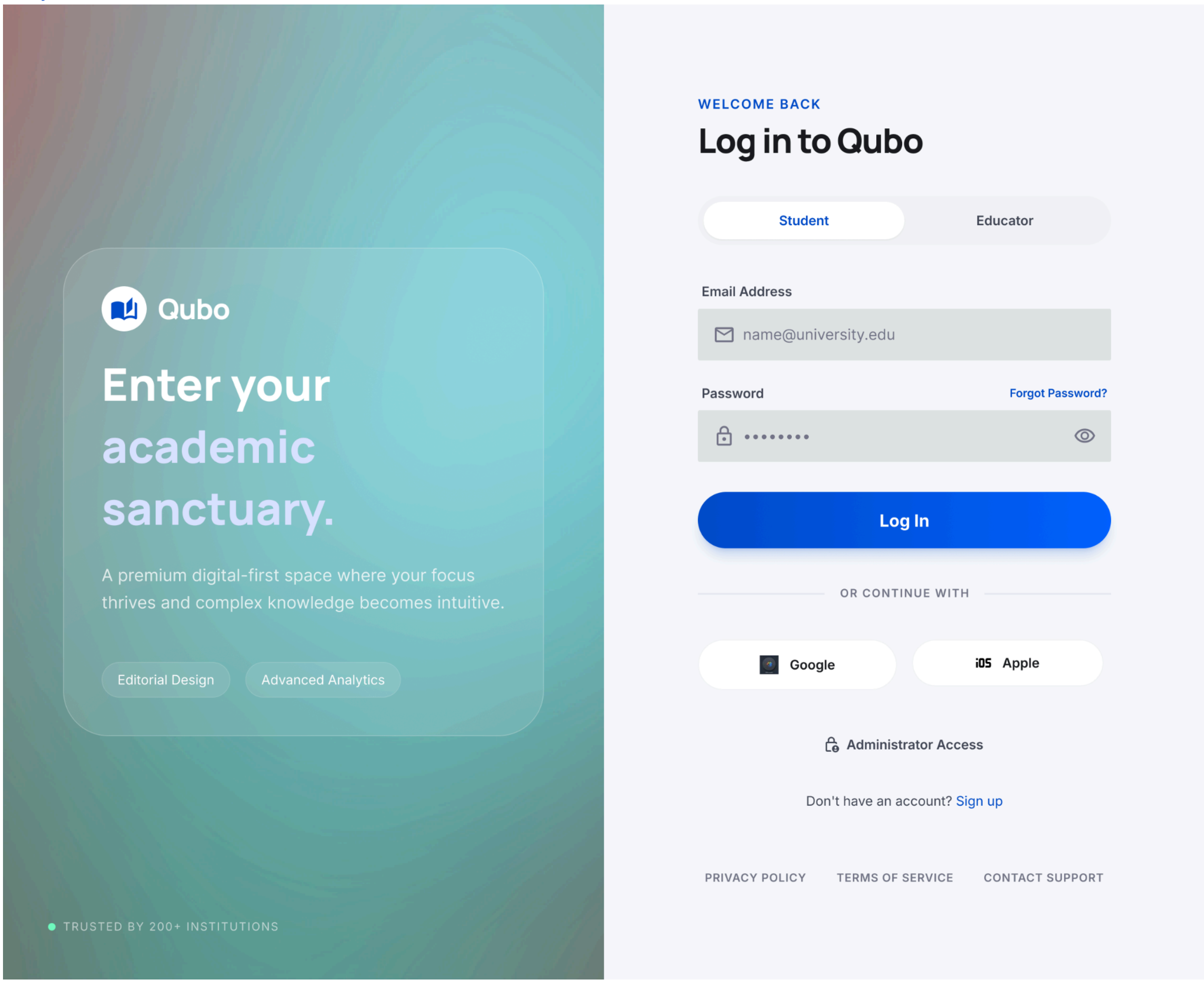


Figure 4.1: Login & Sign Up page

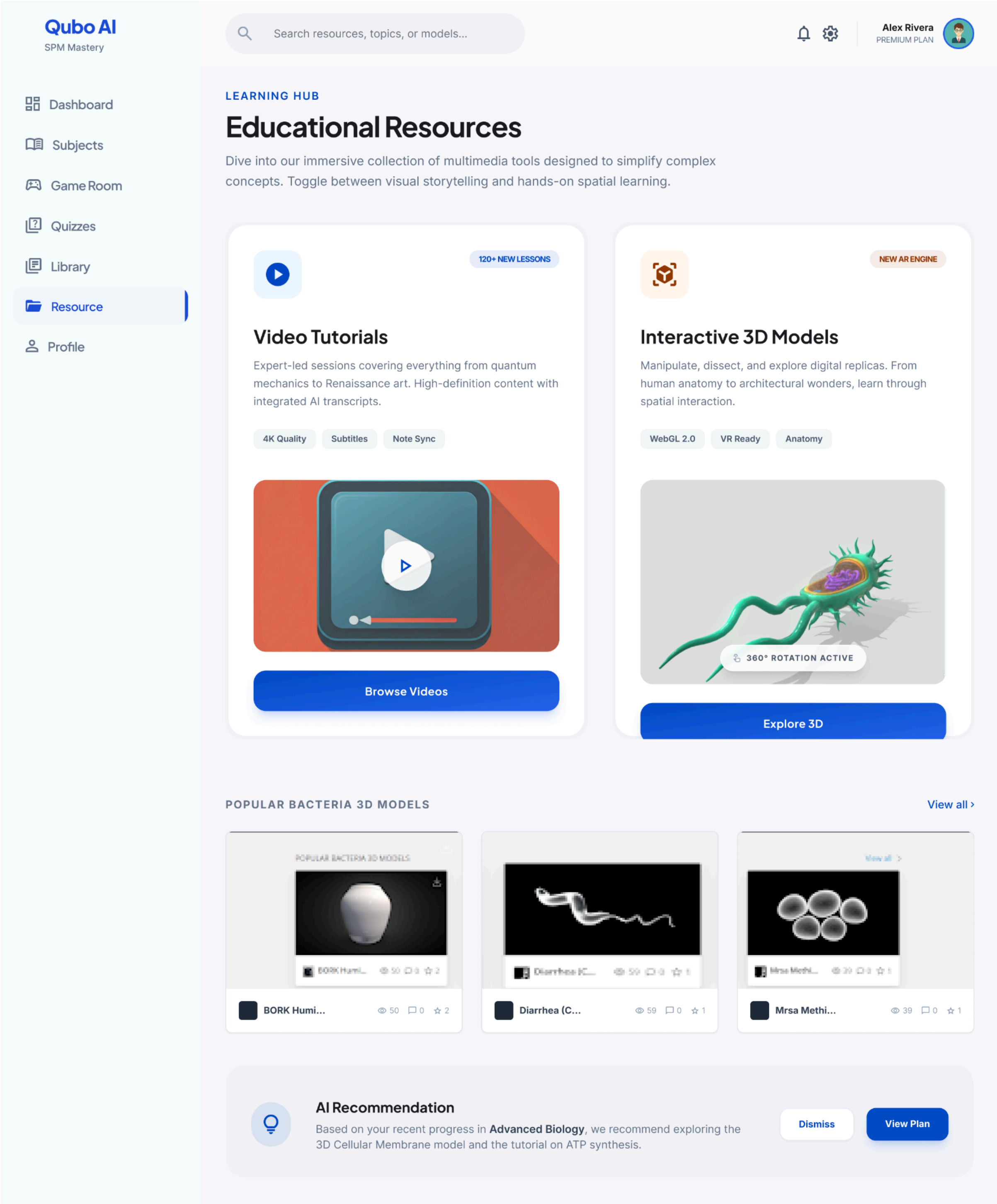


Figure 4.2: Resource Page

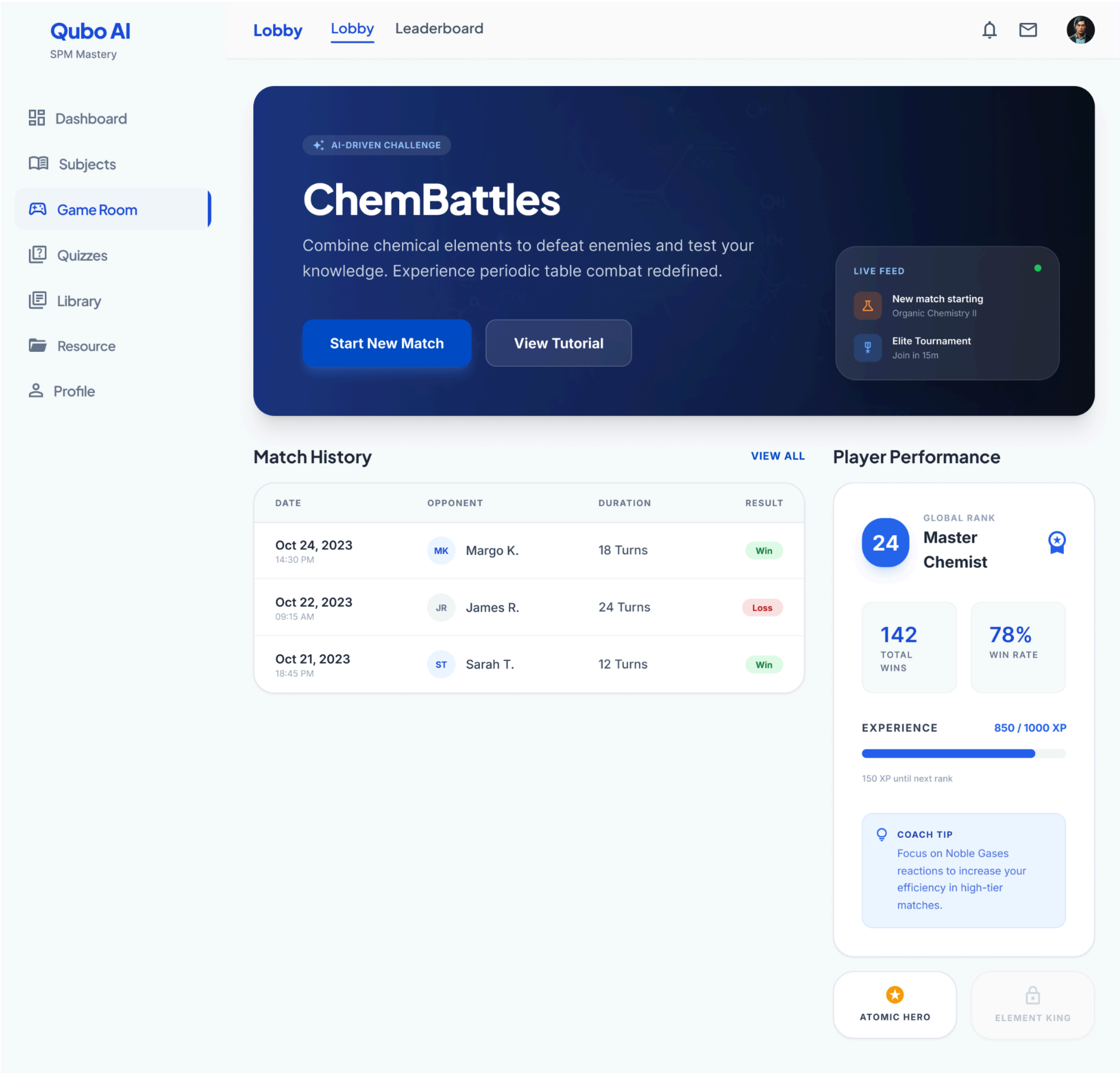


Figure 4.3: Game Room Page

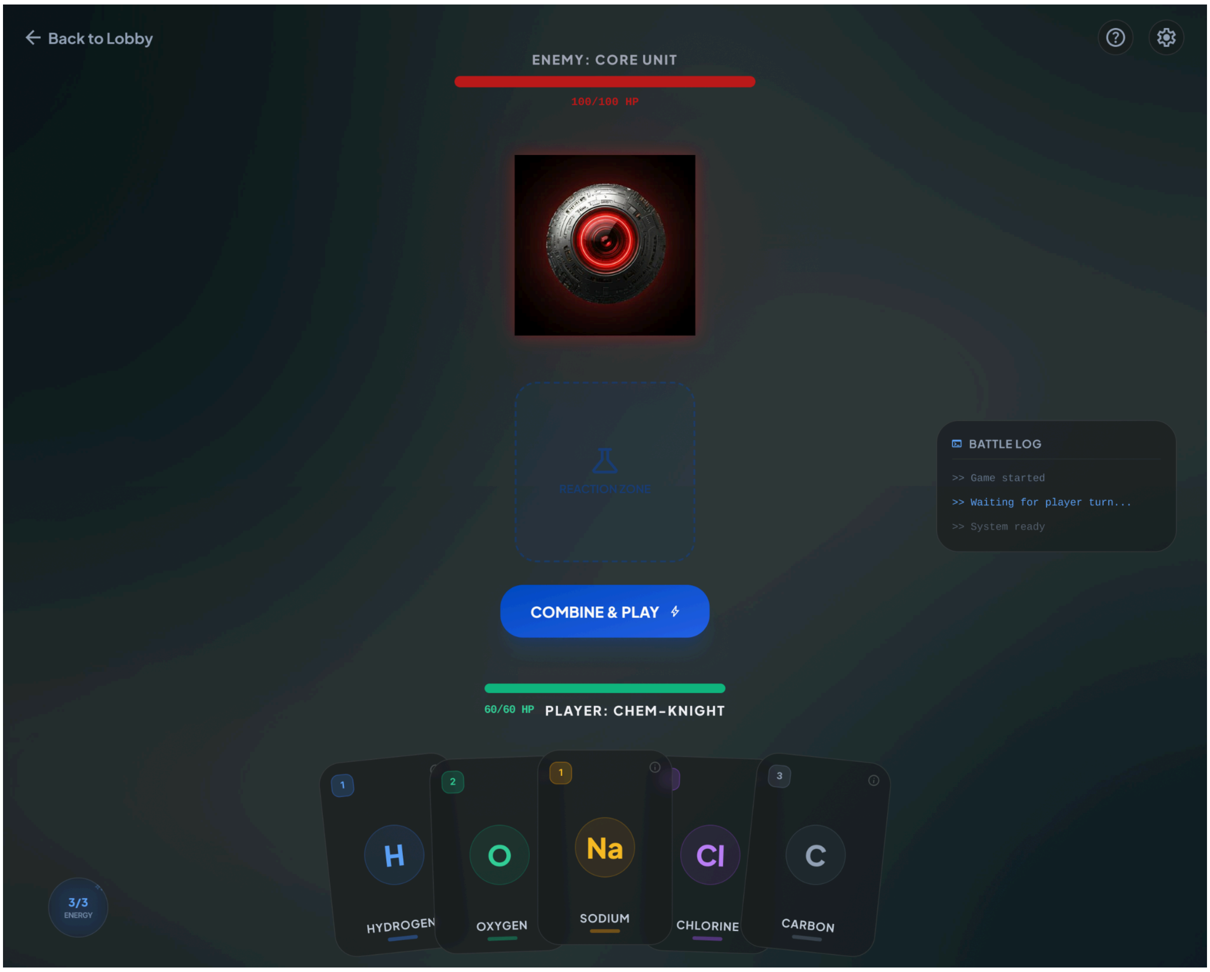


Figure 4.4: Game Page

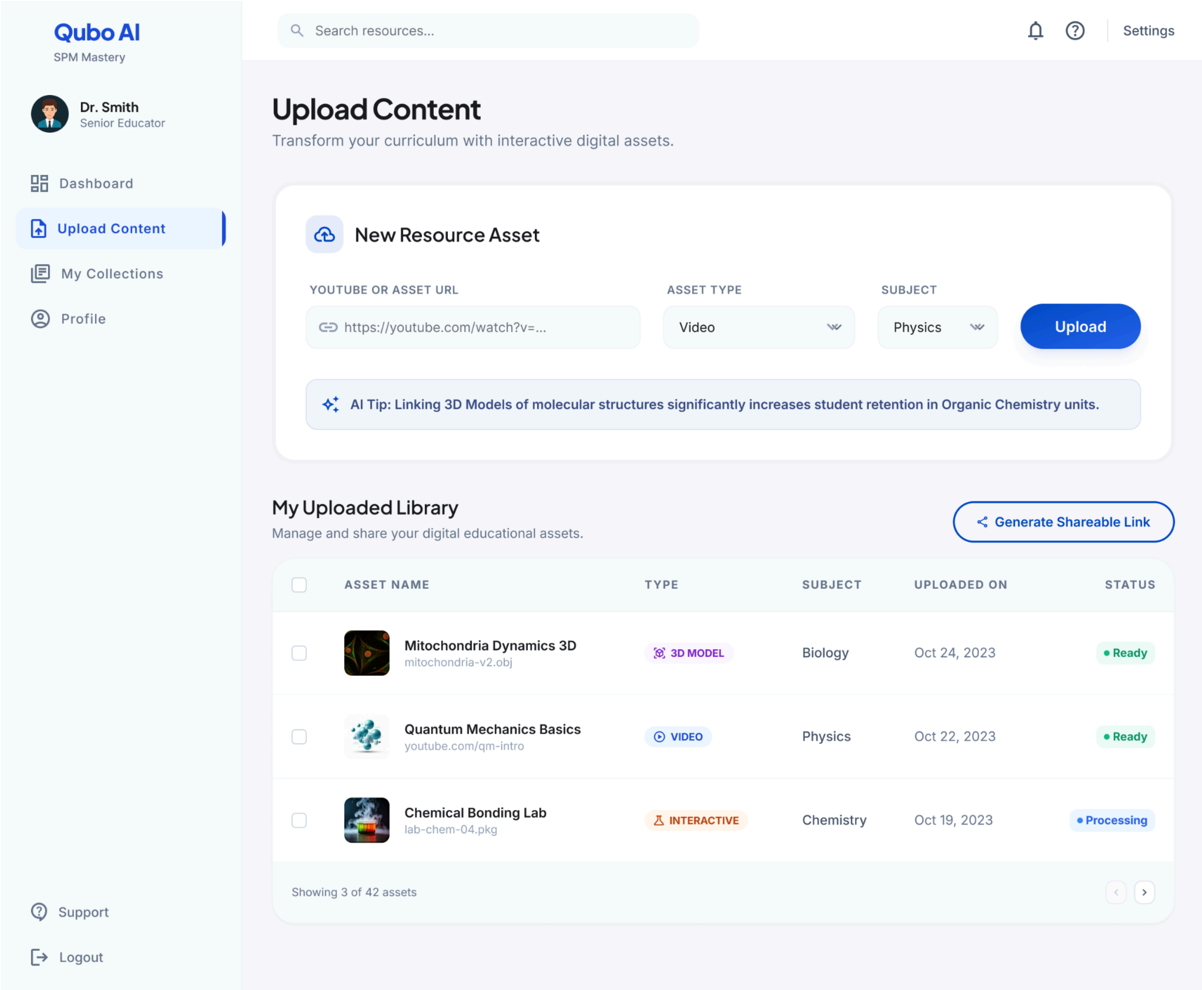


Figure 4.5: Upload Content Page (Educator Dashboard)

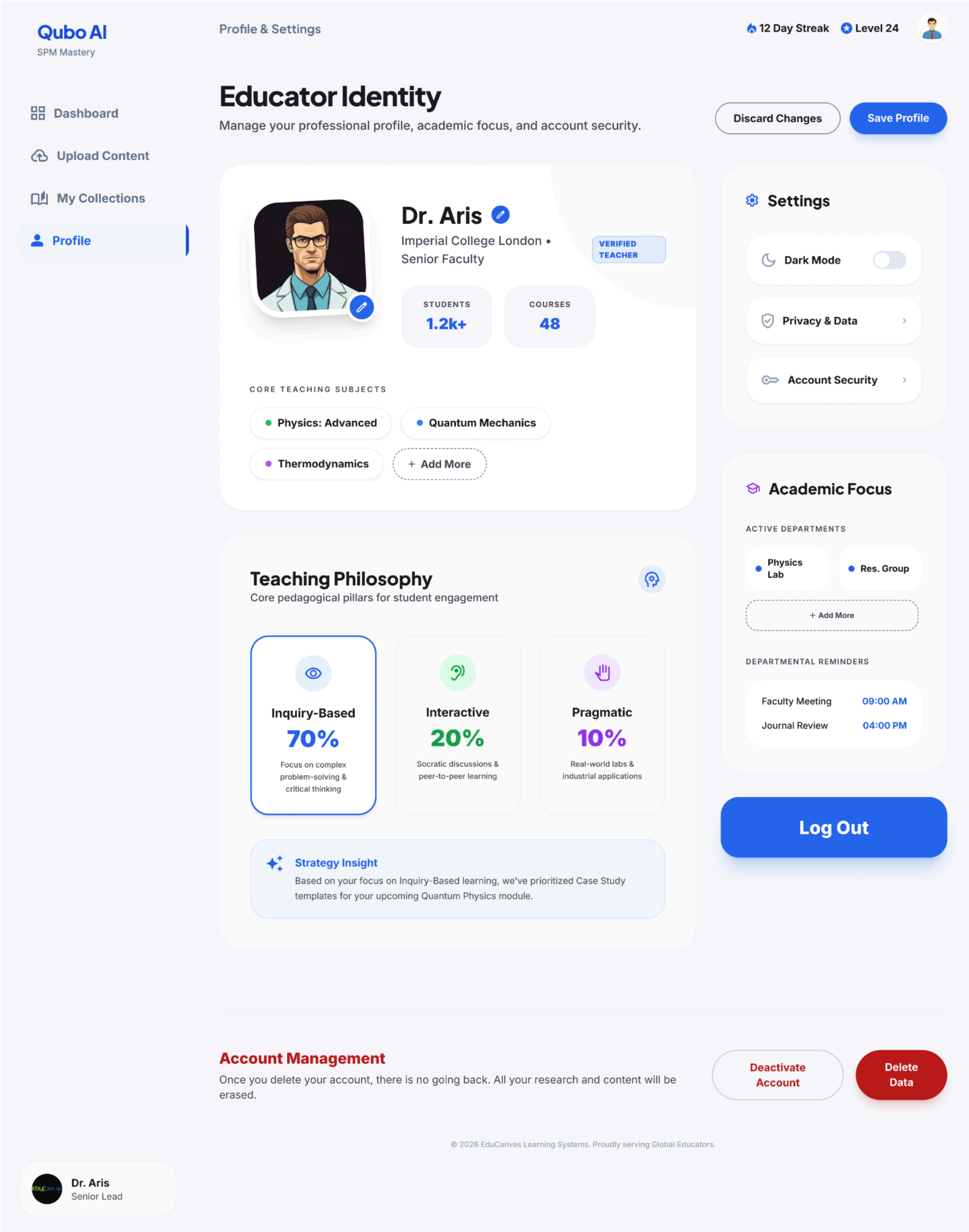


Figure 4.6: Profile Page

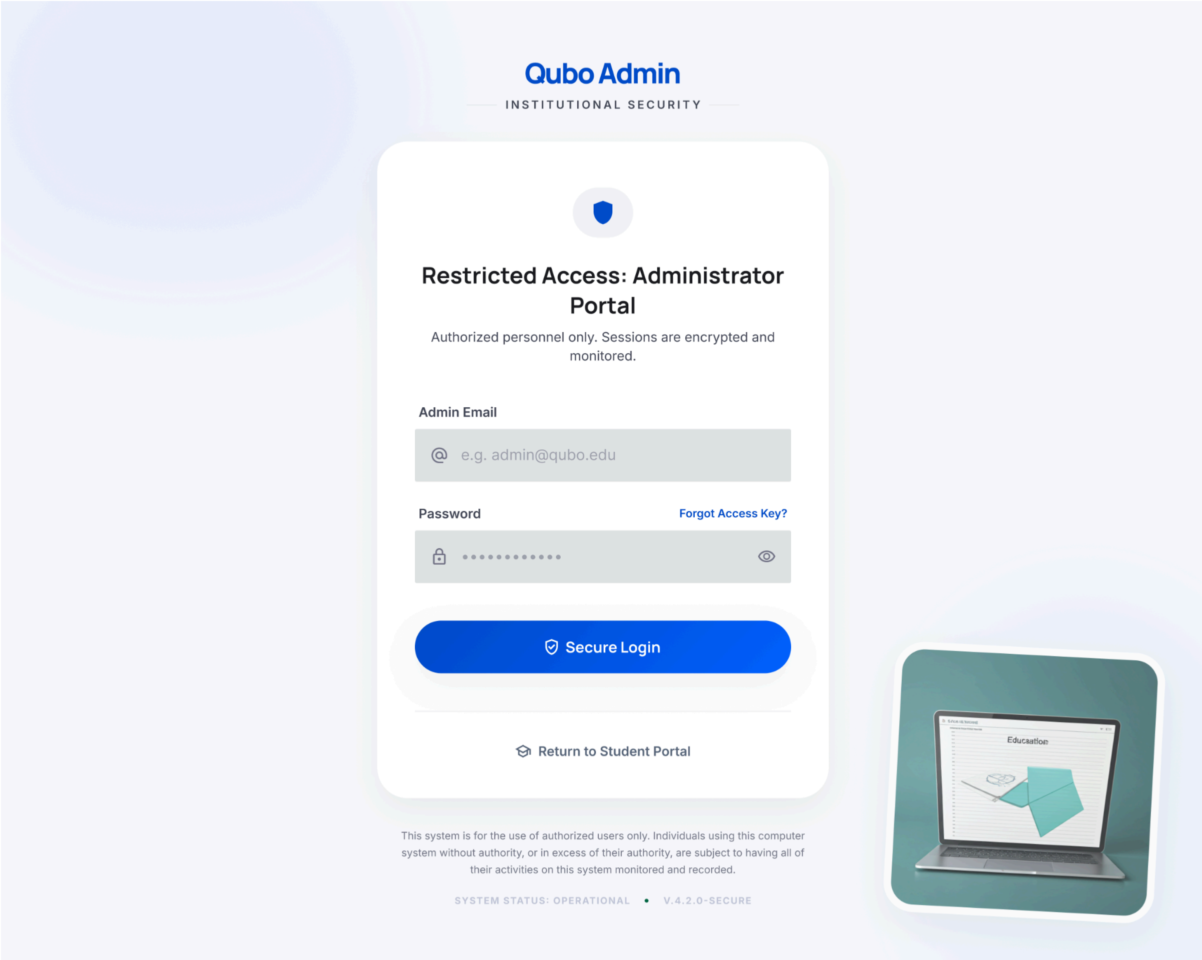


Figure 4.7: Login Page (Admin)

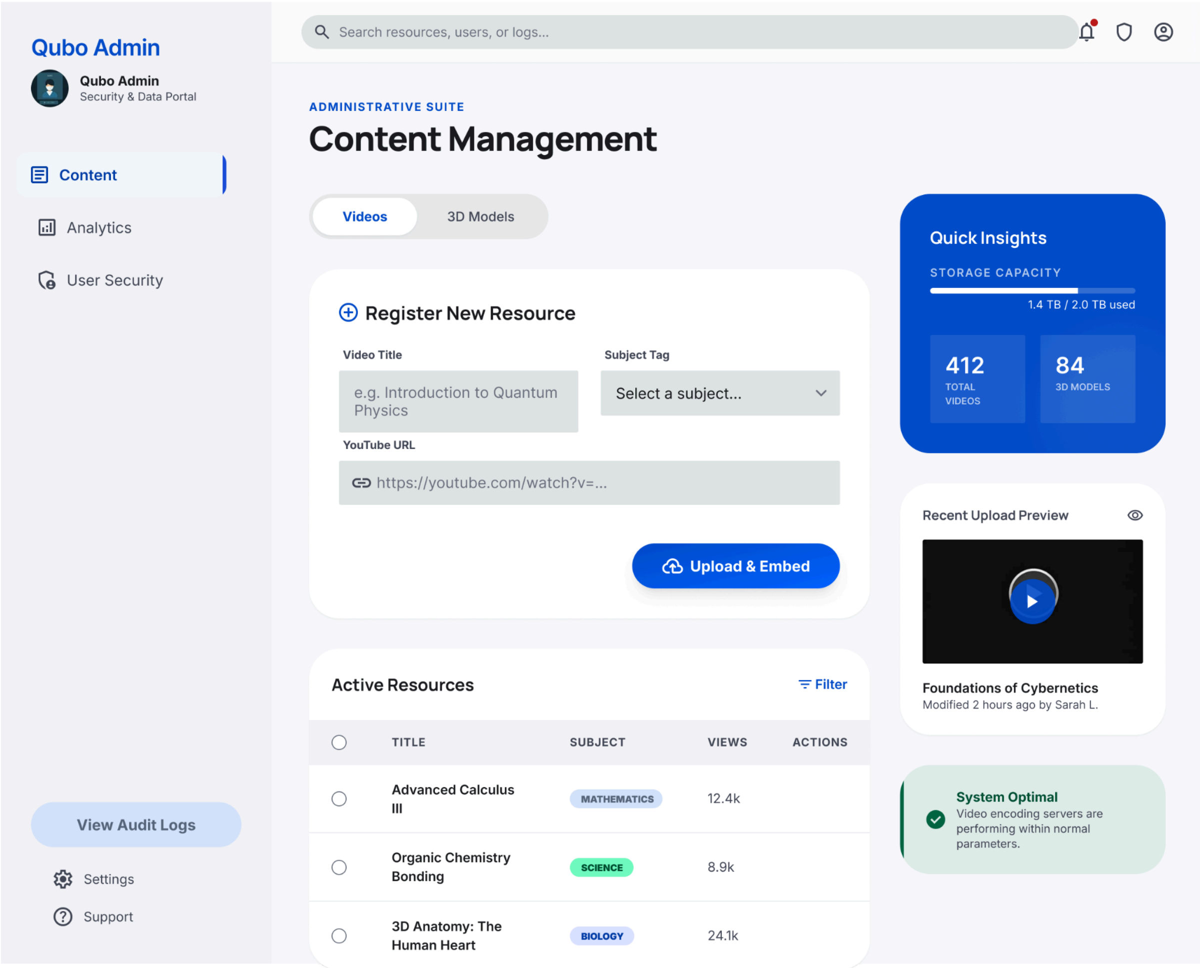


Figure 4.8: Main Page

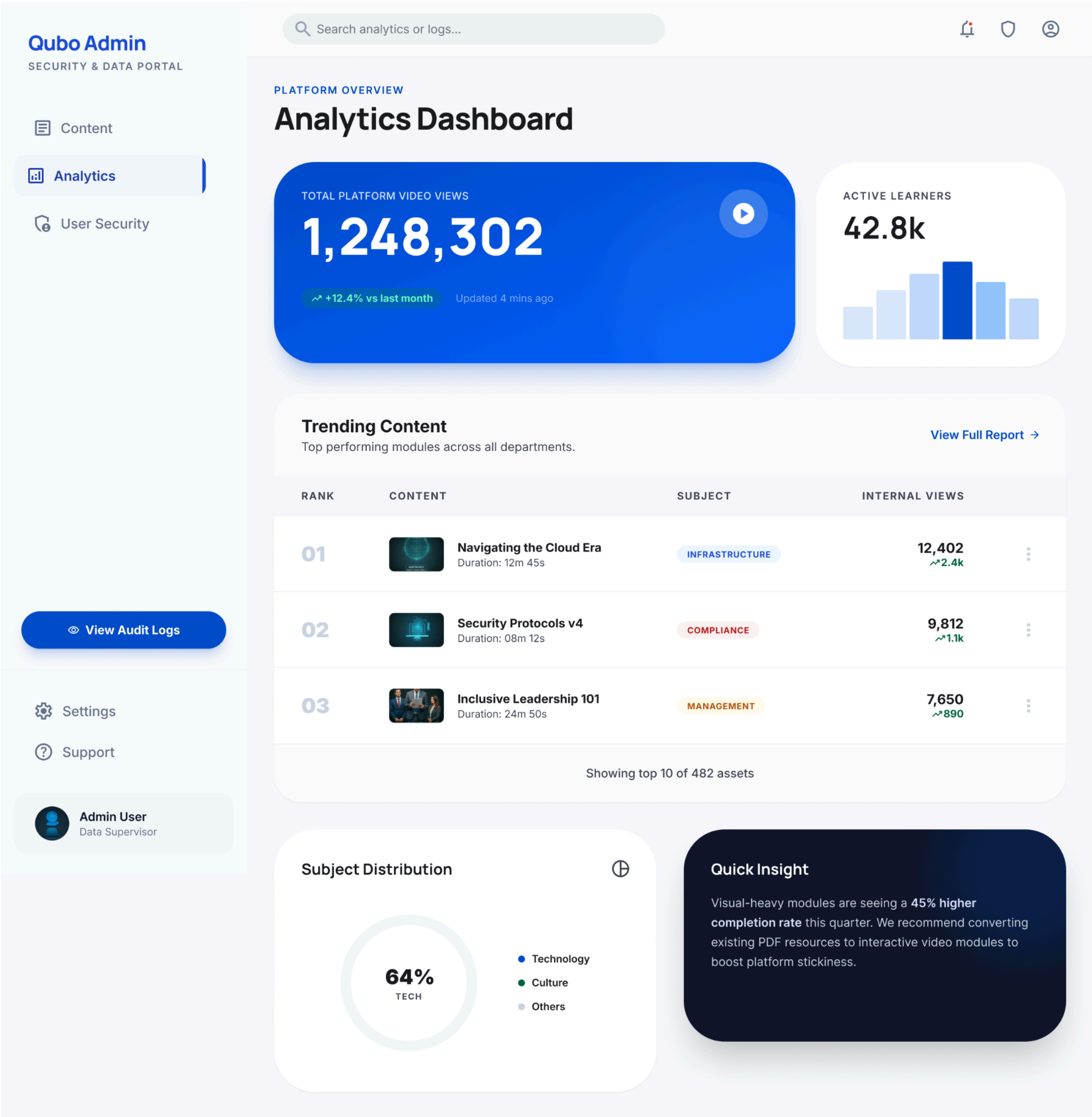


Figure 4.9: Analytics Page (Admin Dashboard)

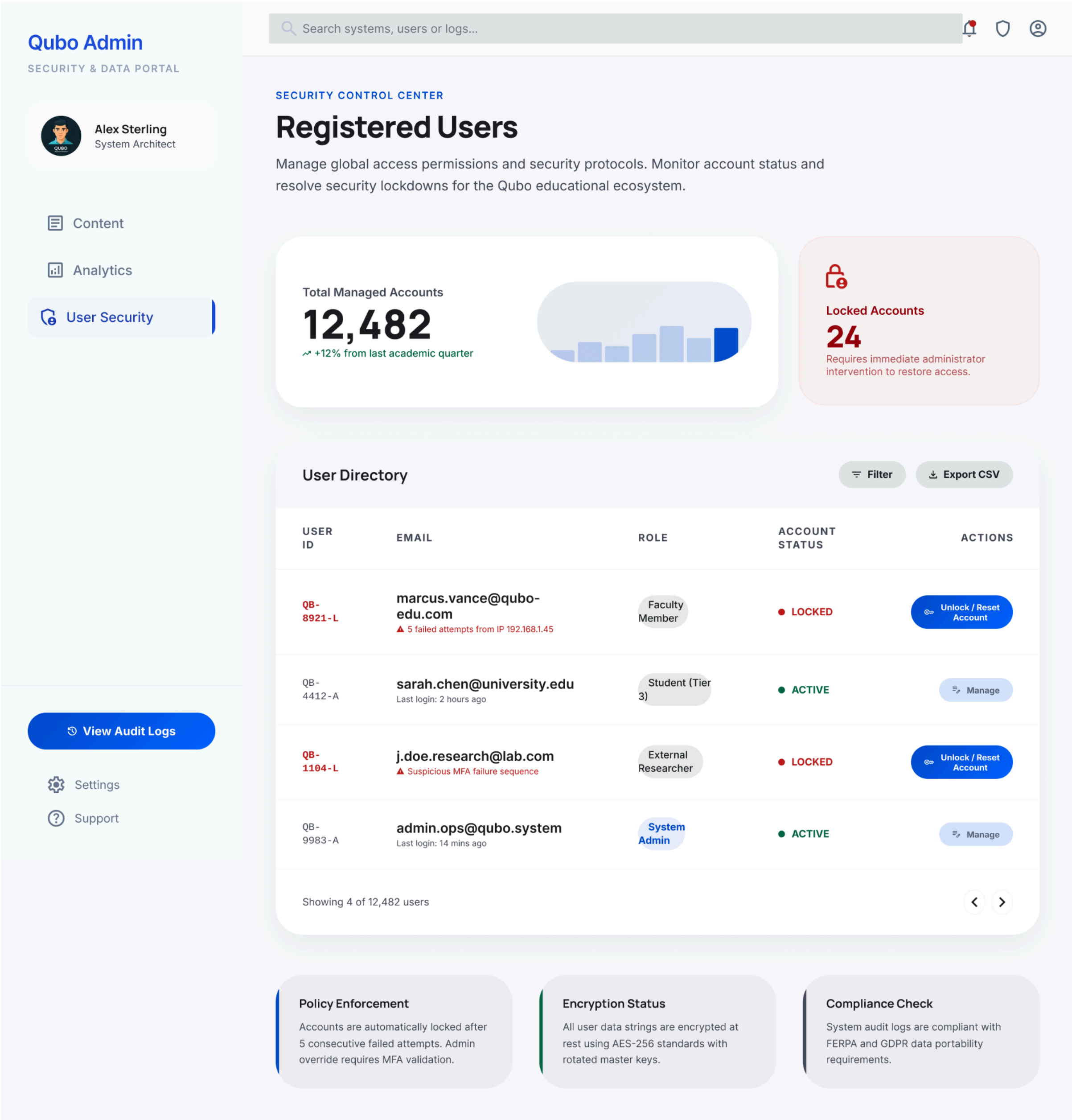


Figure 4.10: User Management Page

4.2 Use case diagram

4.2.1 Personalized Recommendations and Resource Curation Module

https://drive.google.com/file/d/1GcTaE0cD2ib5WEDkfpc1Y3TtTd8jT6mD/view?usp=sharing

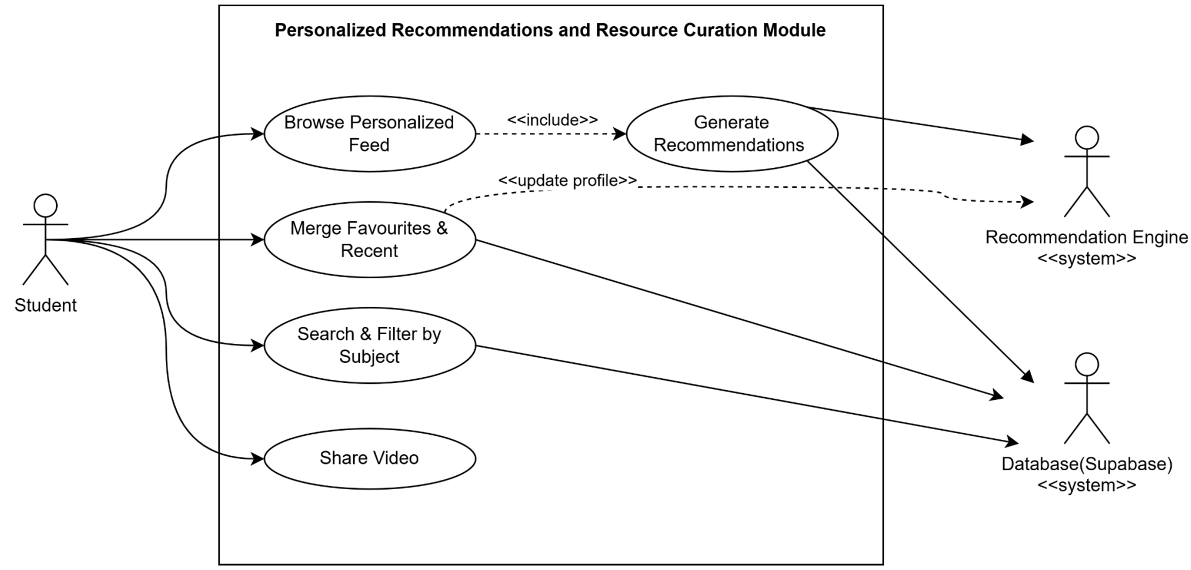


Figure 4.11: Use Case of Personalized Recommendations and Resource Curation Module

4.2.2 Lightweight Gamification Module

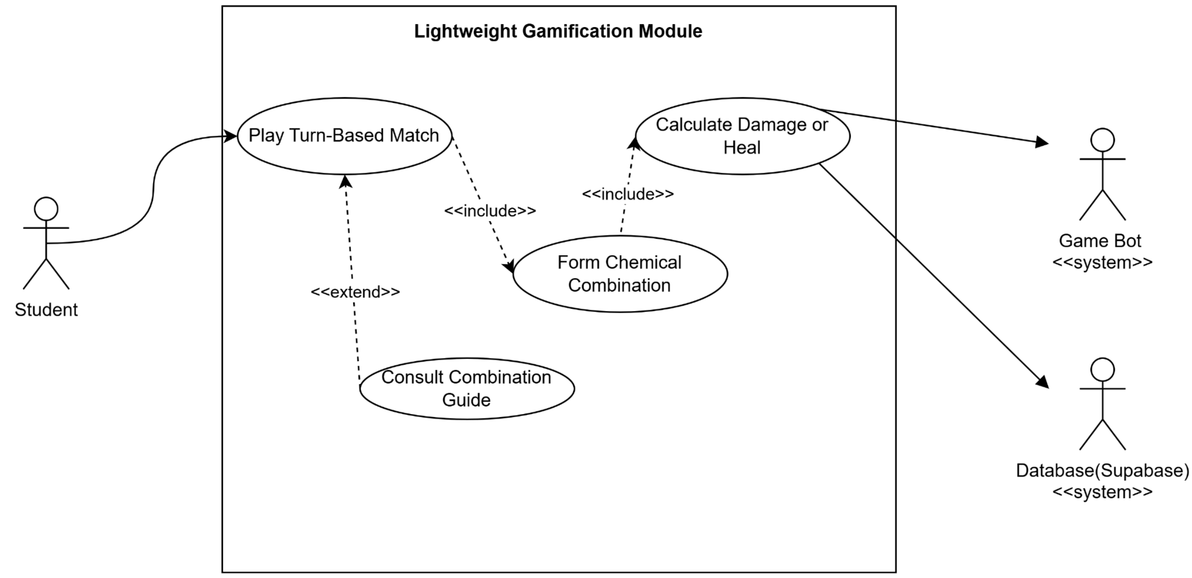


Figure 4.12: Use Case of Lightweight Gamification Module

4.2.3 Resource & System Management Module

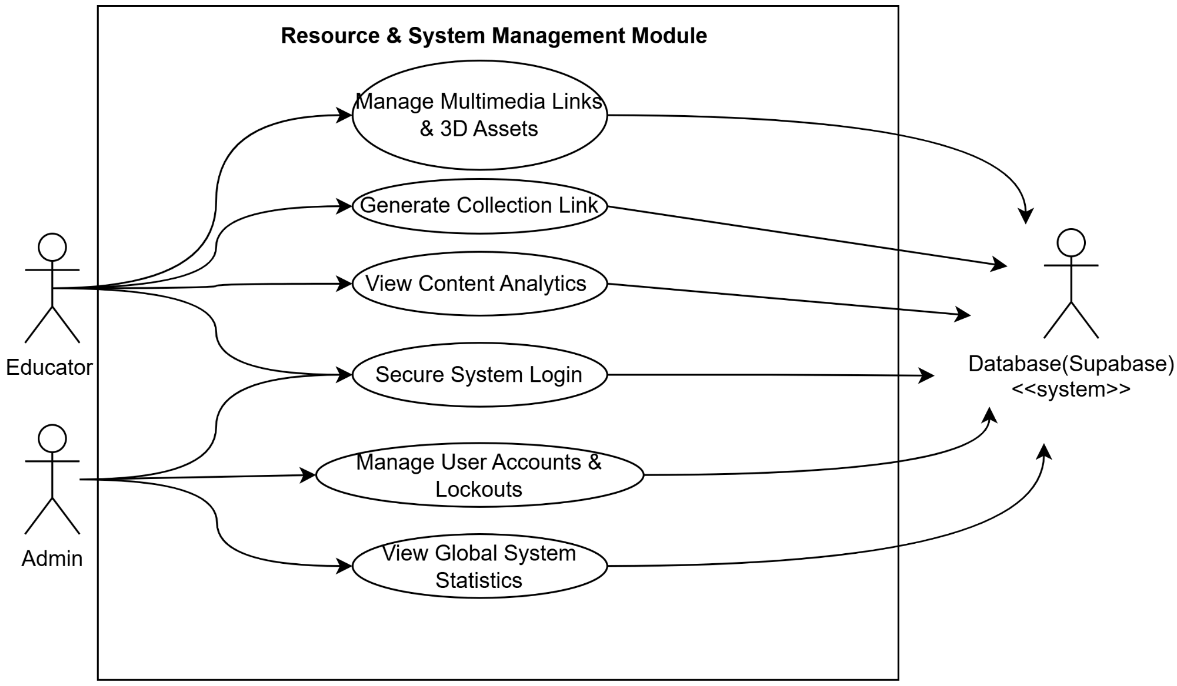


Figure 4.13: Use Case of Resource & System Management Module

4.3 Sequence diagram

4.3.1 Personalized Recommendations and Resource Curation Module

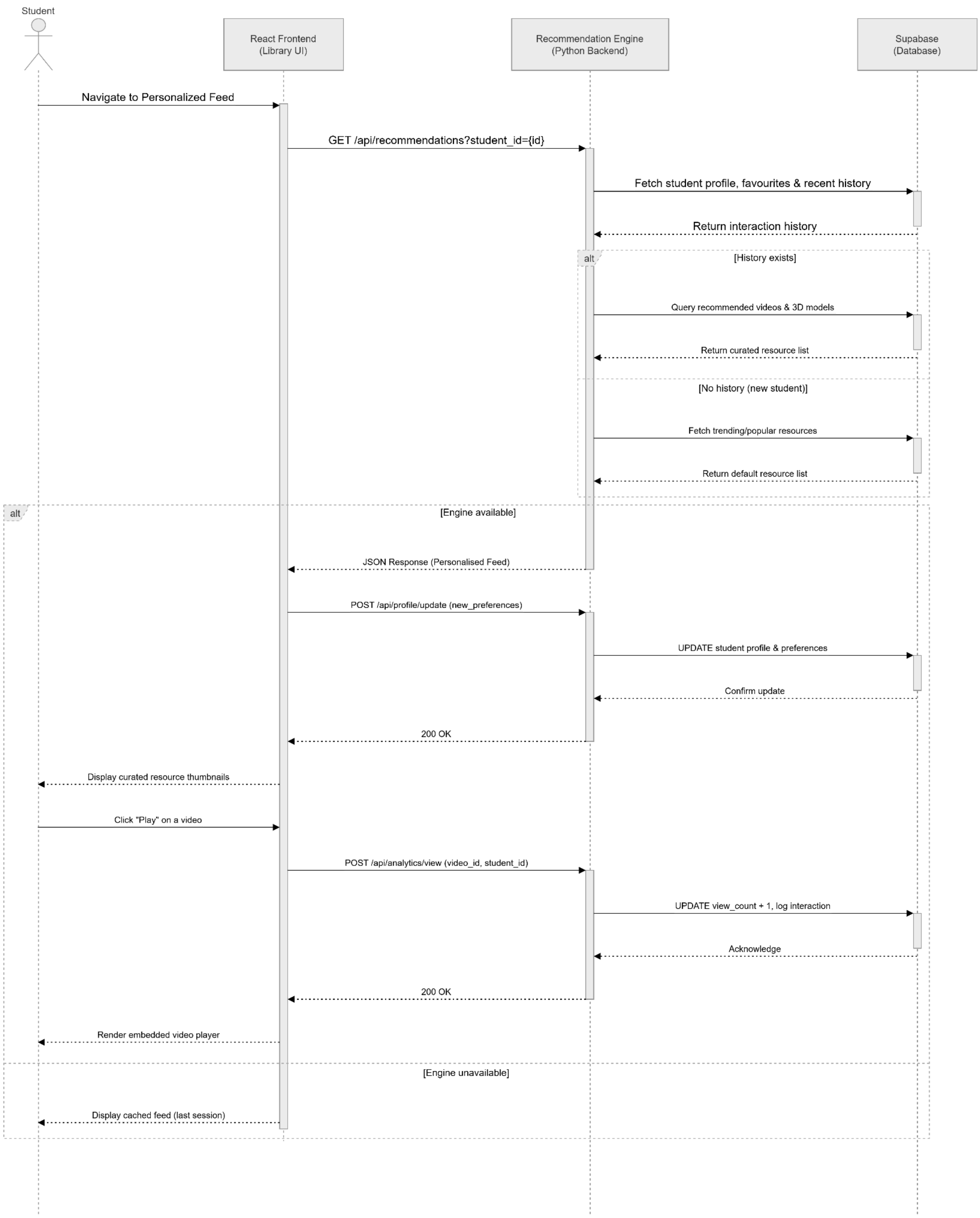
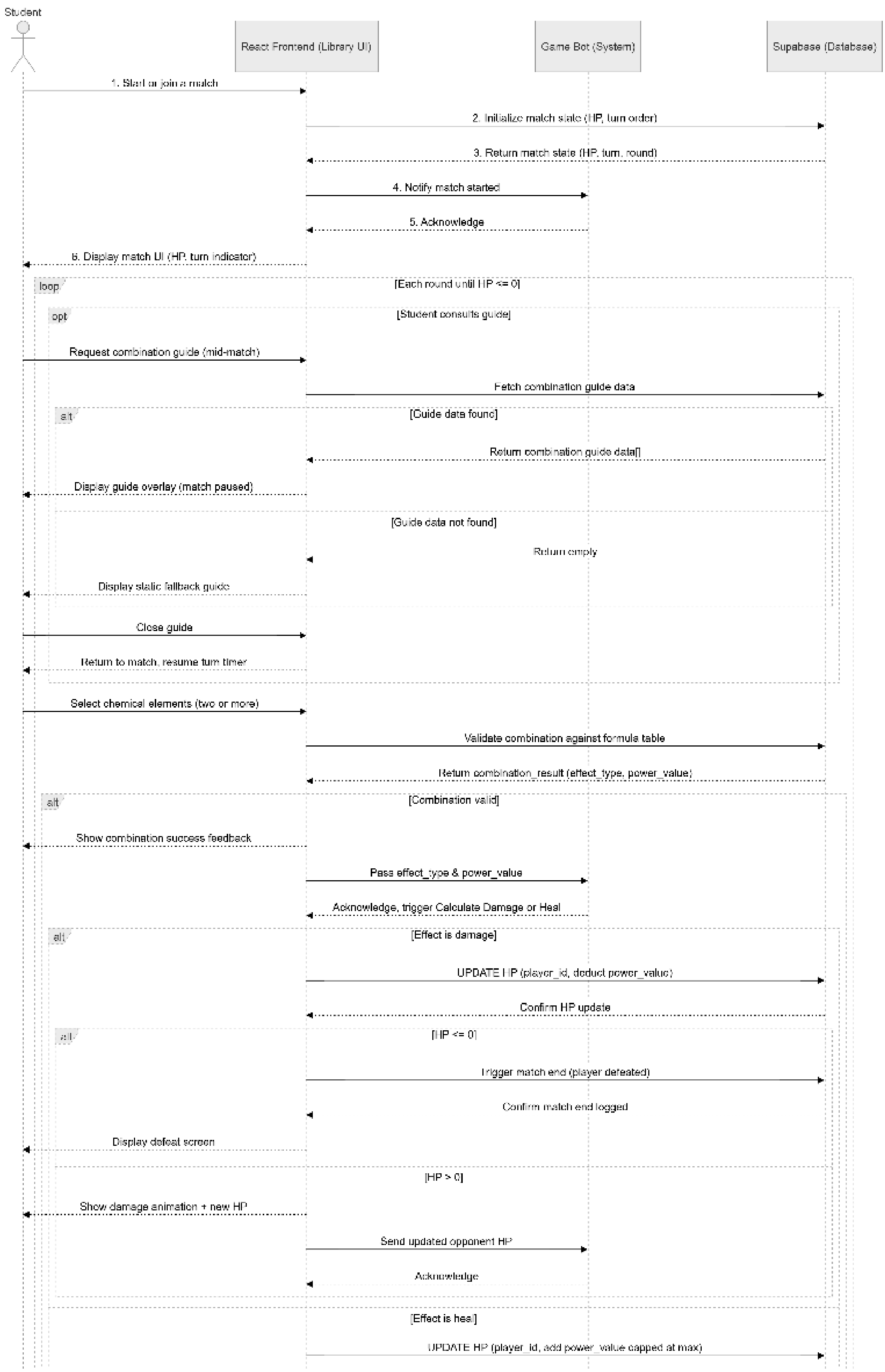


Figure 4.14: Sequence diagram of Personalized Recommendations and Resource Curation Module

4.3.2 Lightweight Gamification Module


Student
React Frontend (Library UI)
Game Bot (System)
Supabase (Database)
1. Start or join a match
2. Initialize match state (HP, turn order)
3. Return match state (HP, turn, round)
4. Notify match started
5. Acknowledge
6. Display match UI (HP, turn indicator)
loop
[Each round until HP <= 0]
opt
[Student consults guide]
Request combination guide (mid-match)
Fetch combination guide data
alt
[Guide data found]
Return combination guide data[]
Display guide overlay (match paused)
[Guide data not found]
Return empty
Display static fallback guide
Close guide
Return to match, resume turn timer
Select chemical elements (two or more)
Validate combination against formula table
Return combination_result (effect_type, power_value)
alt
[Combination valid]
Show combination success feedback
Pass effect_type & power_value
Acknowledge, trigger Calculate Damage or Heal
alt
[Effect is damage]
UPDATE HP (player_id, deduct power_value)
Confirm HP update
alt
[HP <= 0]
Trigger match end (player defeated)
Confirm match end logged
Display defeat screen
[HP > 0]
Show damage animation + new HP
Send updated opponent HP
Acknowledge
[Effect is heal]
UPDATE HP (player_id, add power_value capped at max)

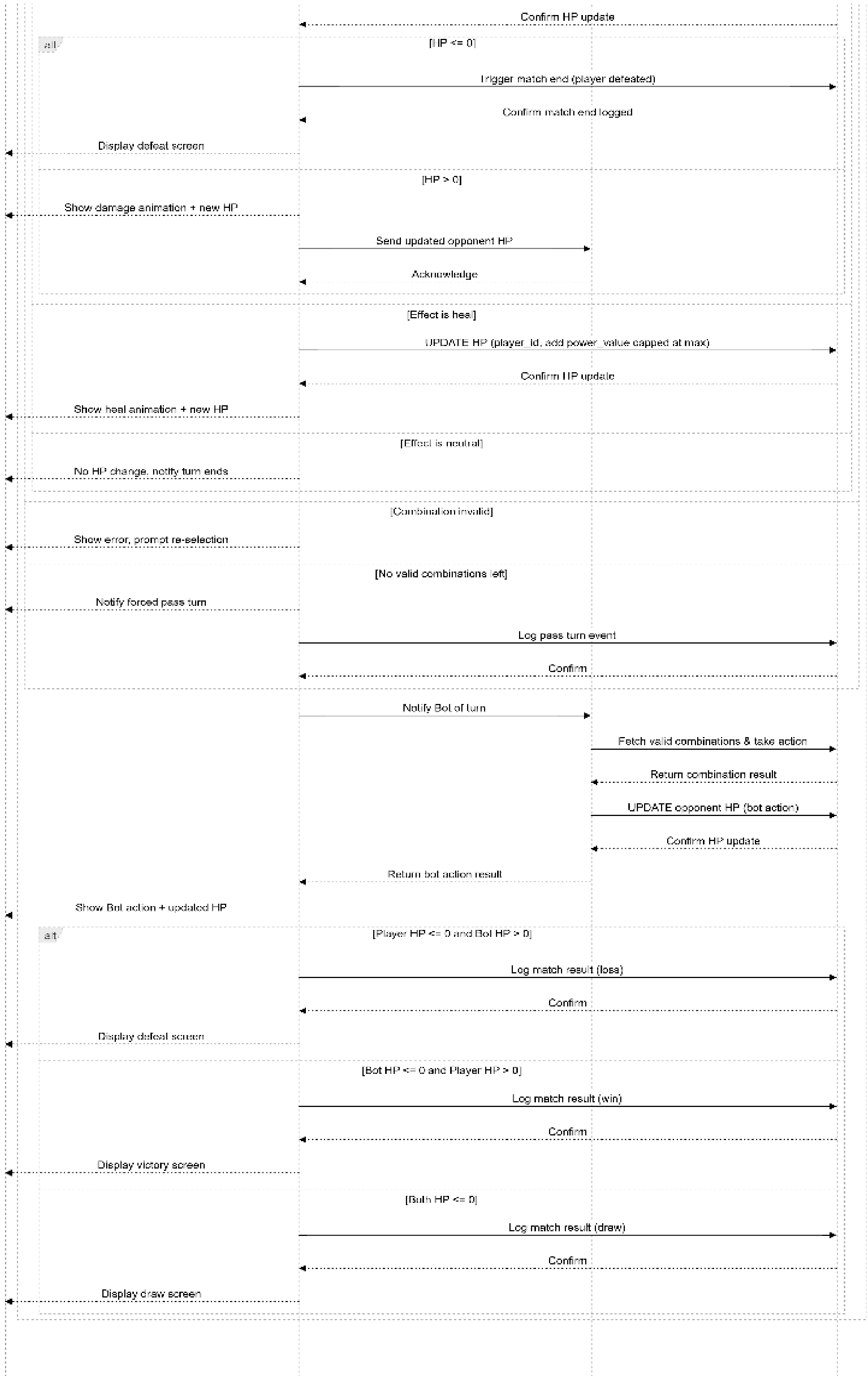


Figure 4.15: Sequence Diagram of Lightweight Gamification Module

4.3.2.1 Lightweight Gamification Module(Play Turn-Based Match)

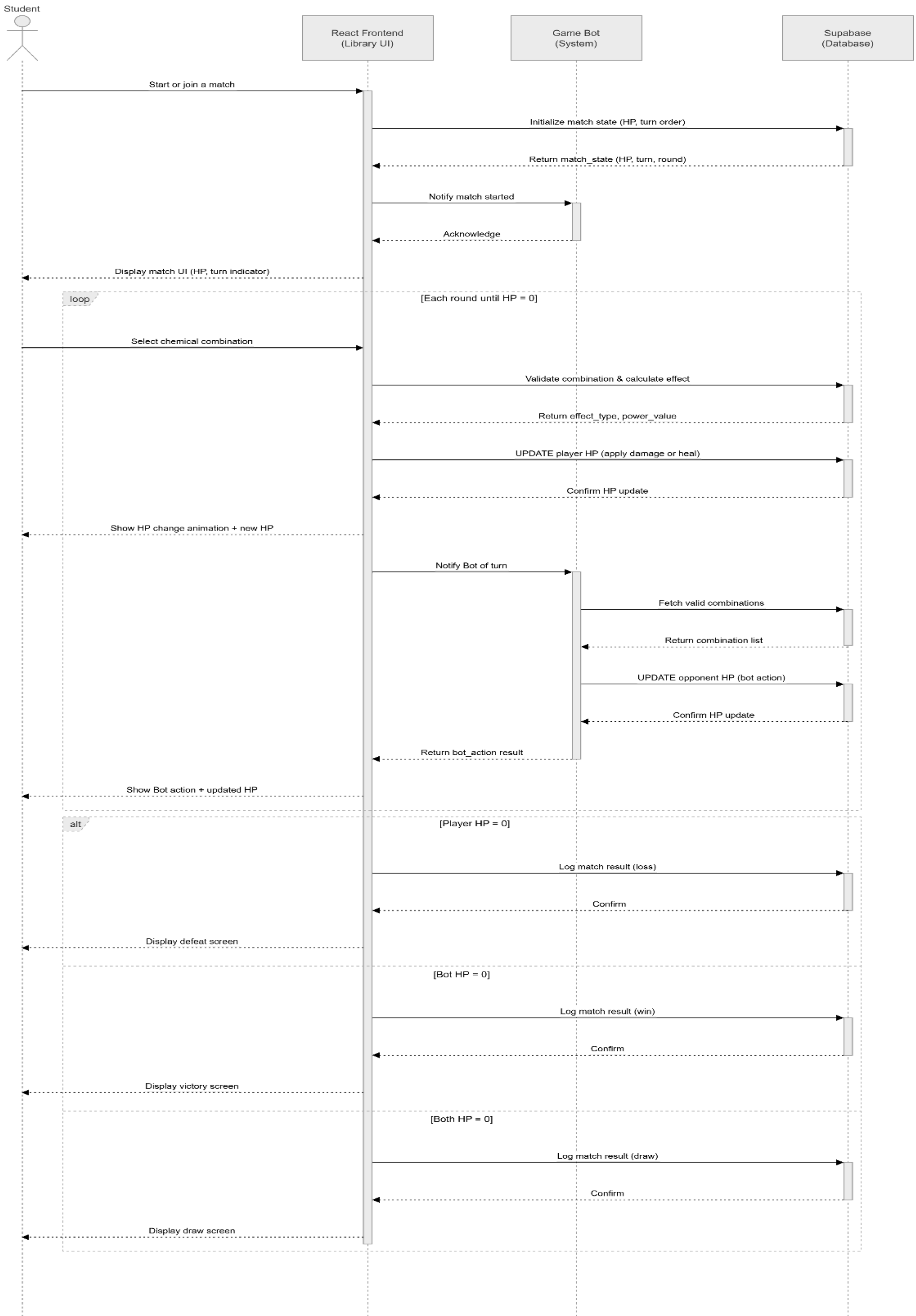


Figure 4.16: Sequence Diagram of Lightweight Gamification Module (Play Turn-Based Match)

4.3.2.2 Lightweight Gamification Module(Form Chemical Combination)

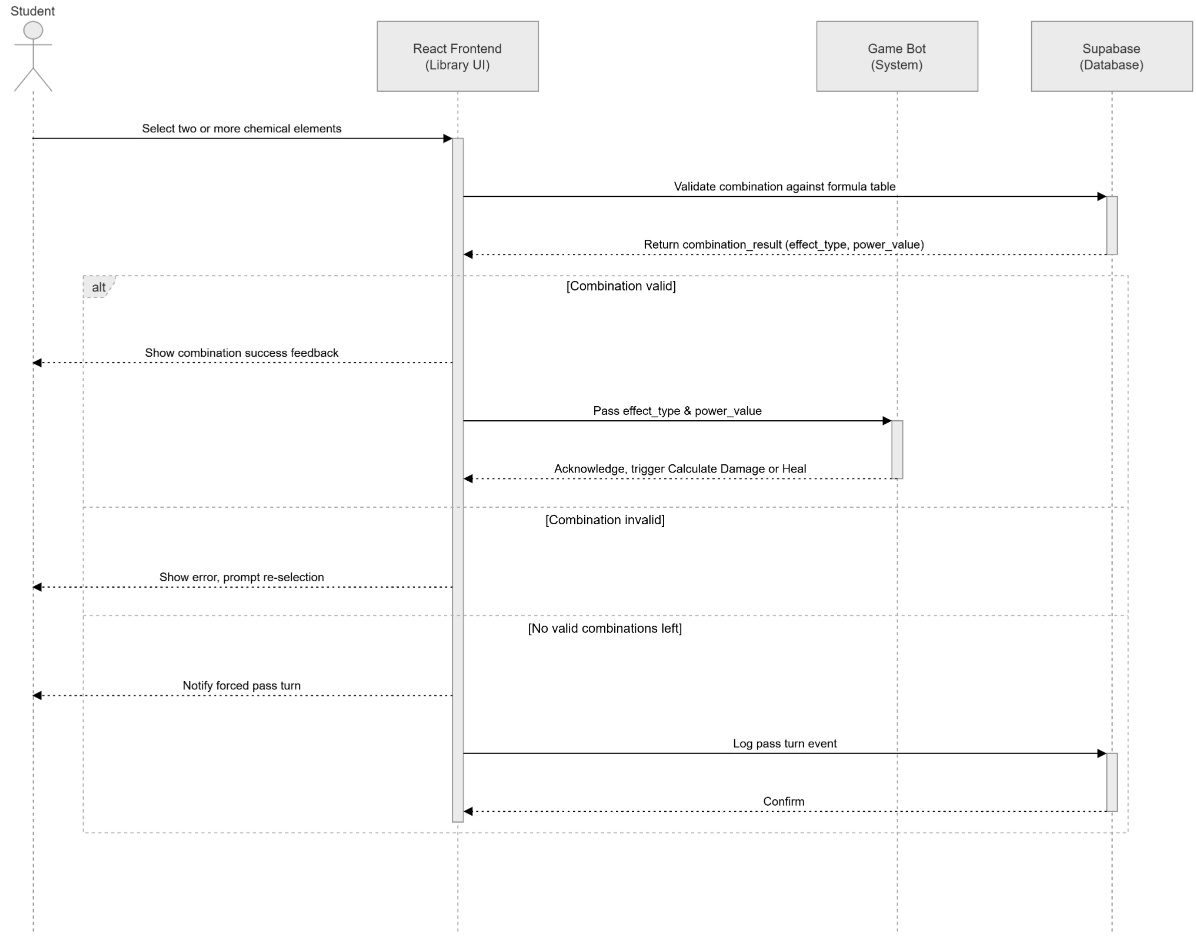


Figure 4.17: Sequence Diagram of Lightweight Gamification Module (Form Chemical Combination)

4.3.2.3 Lightweight Gamification Module(Calculate Damage or Heal)

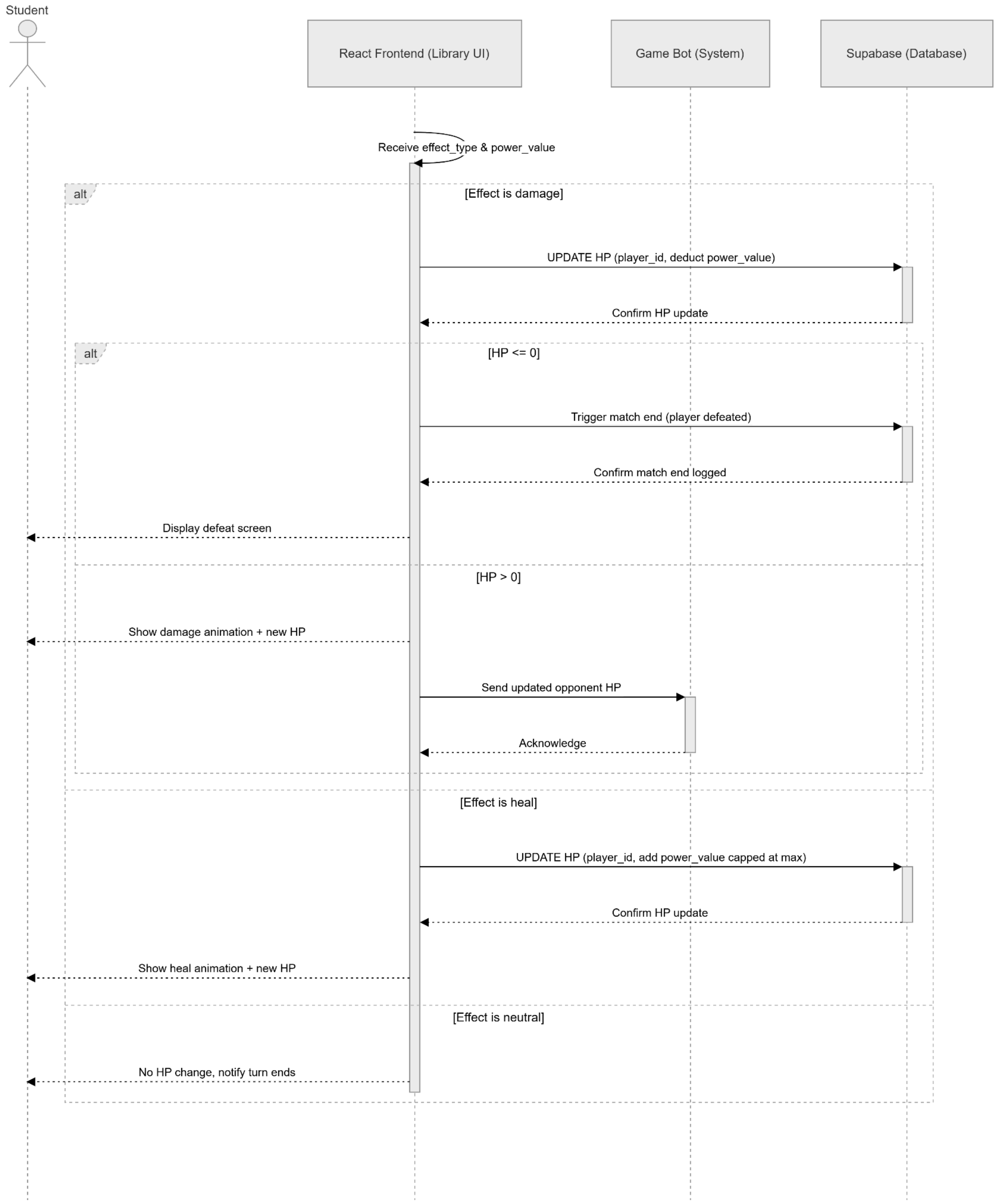


Figure 4.18: Sequence Diagram of Lightweight Gamification Module (Calculate Damage or Heal)

4.3.2.4 Lightweight Gamification Module(Consult Combination Guide)

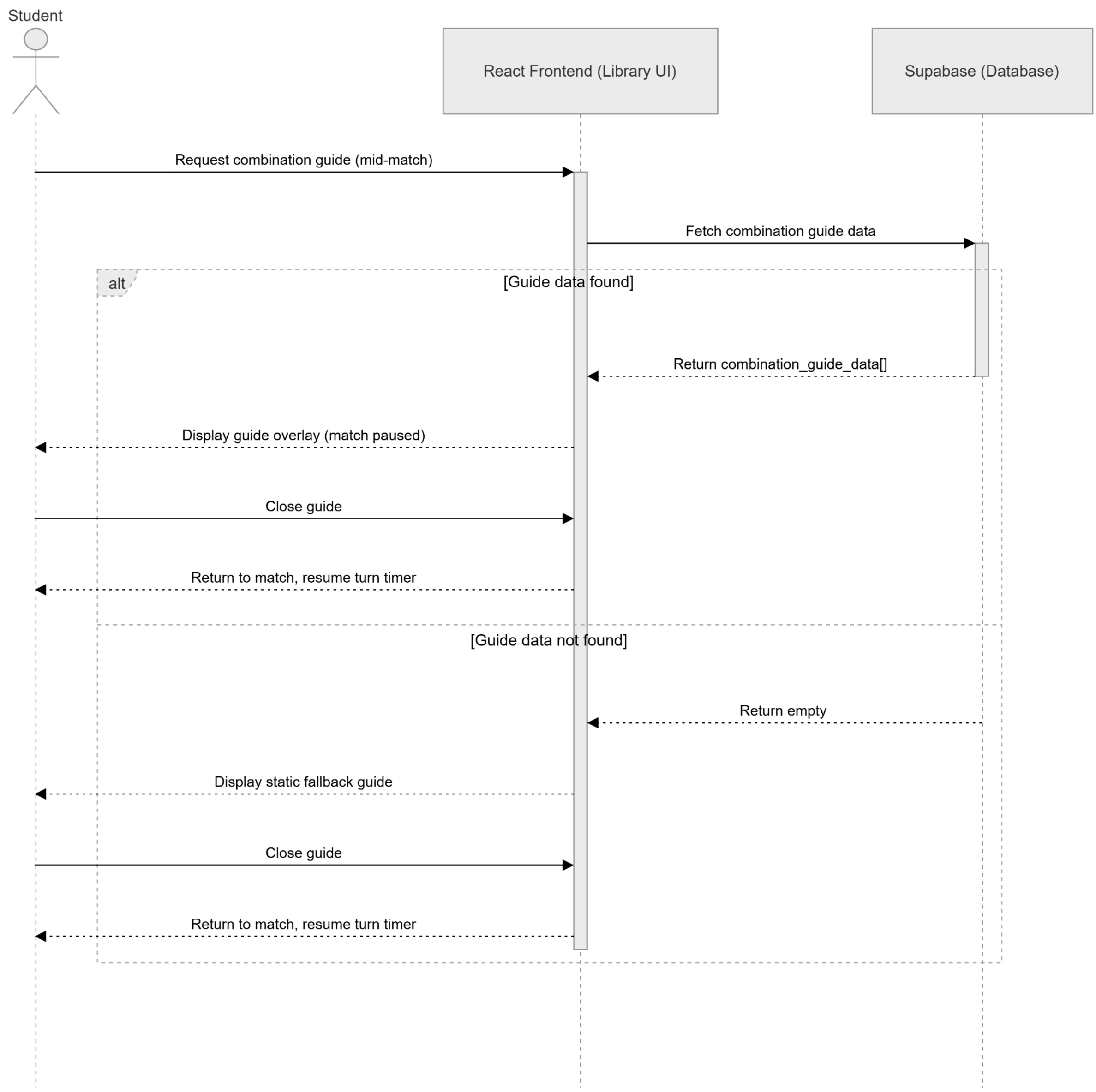


Figure 4.19: Sequence Diagram of Lightweight Gamification Module(Consult Combination Guide)

4.3.3 Resource & System Management Module

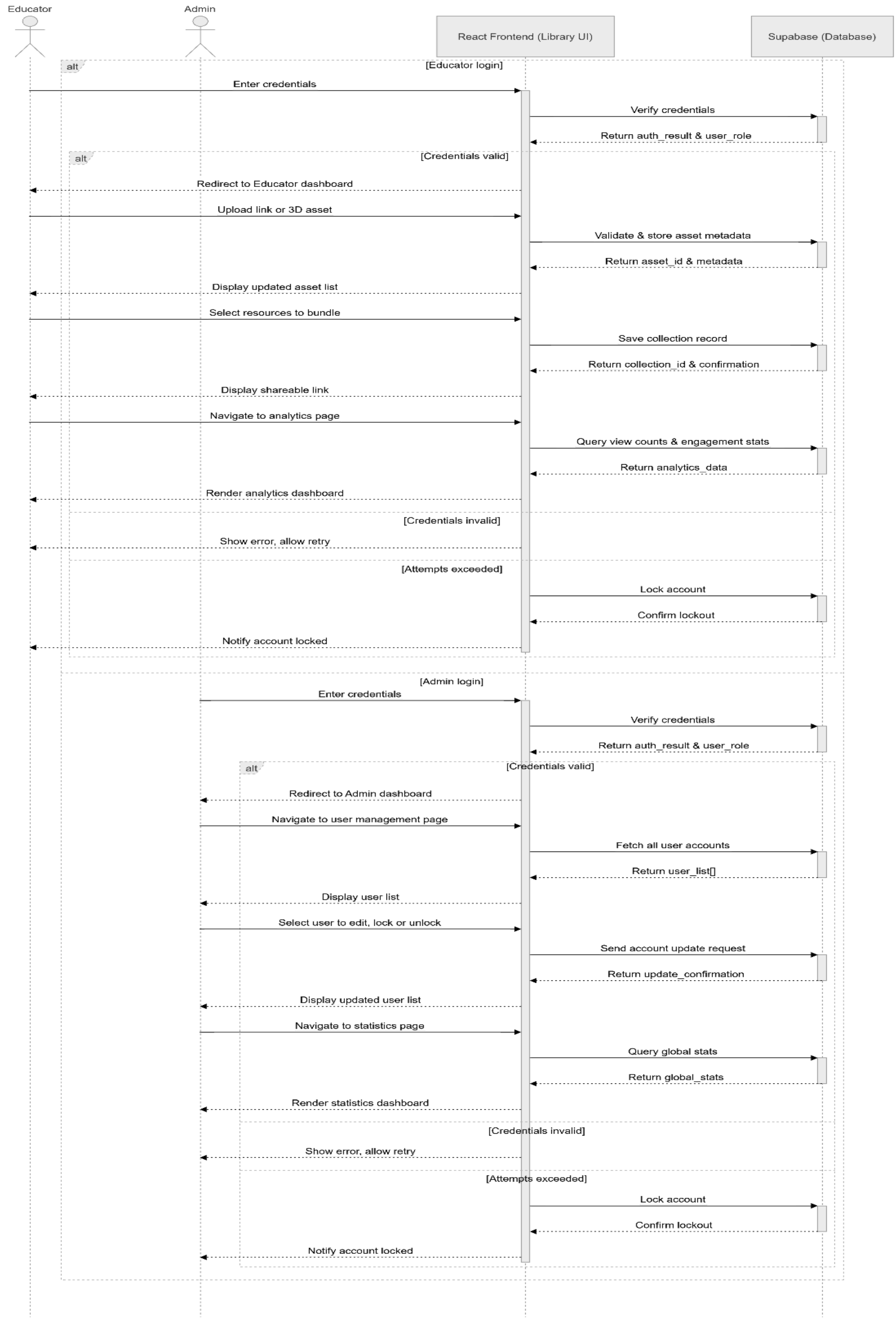


Figure 4.20: Sequence Diagram of Resource & System Management Module

4.3.3.1 Resource & System Management(Educator Flows)

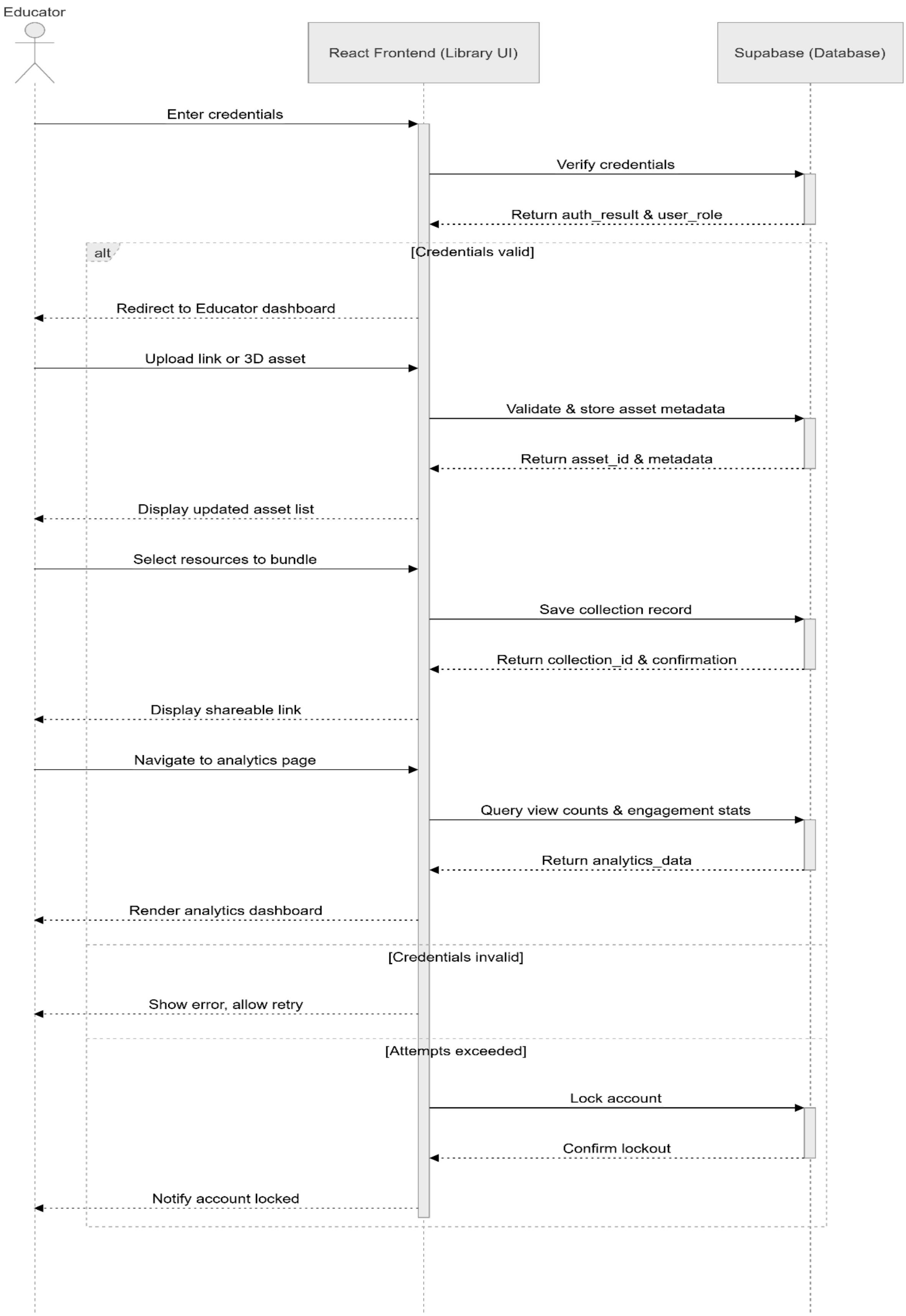


Figure 4.21: Sequence Diagram of Resource & System Management(Educator Flows)

4.3.3.2 Resource & System Management(Admin Flows)

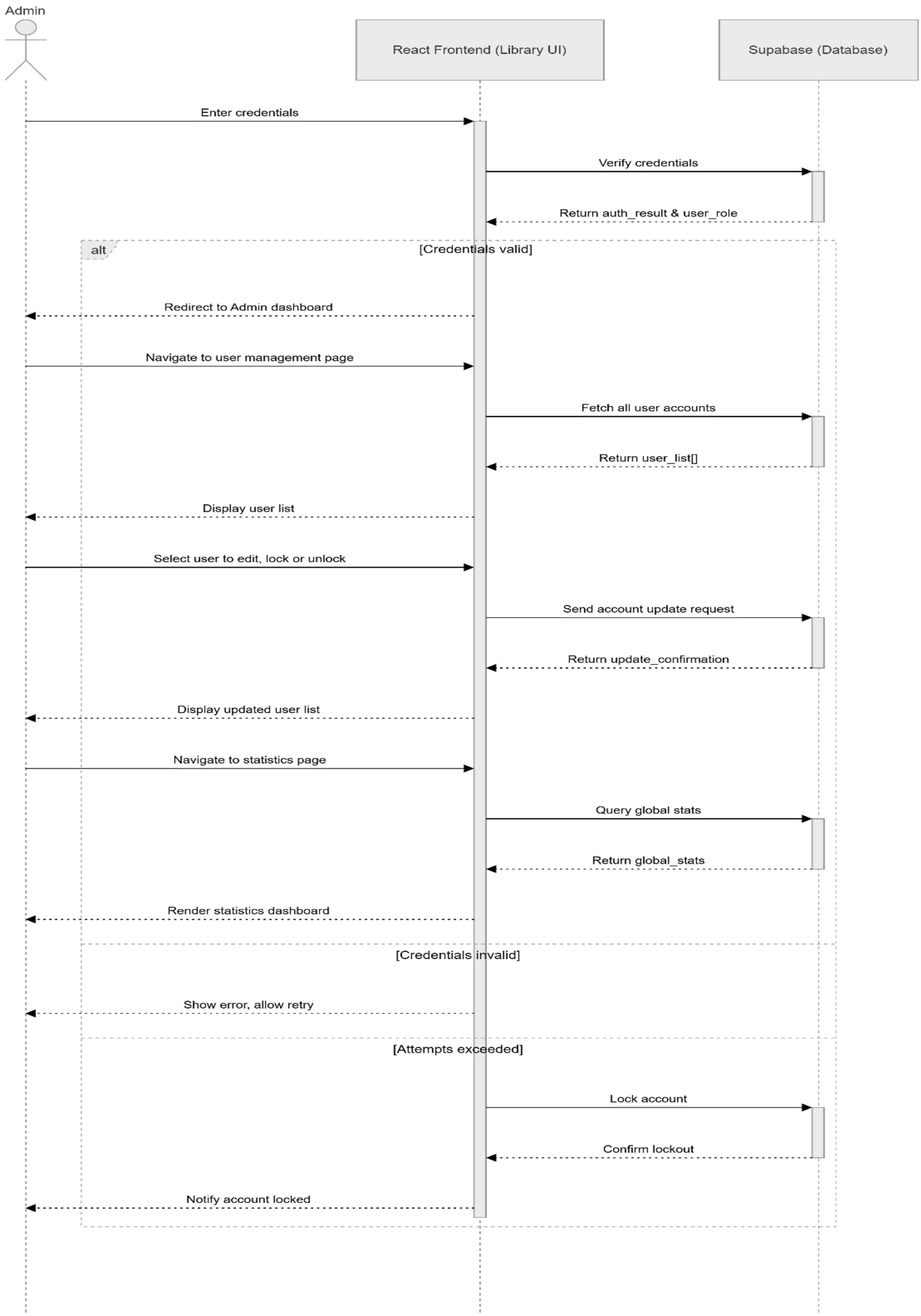


Figure 4.22: Sequence Diagram of Resource & System Management(Admin Flows)

4.4 Entity Relationship Diagram

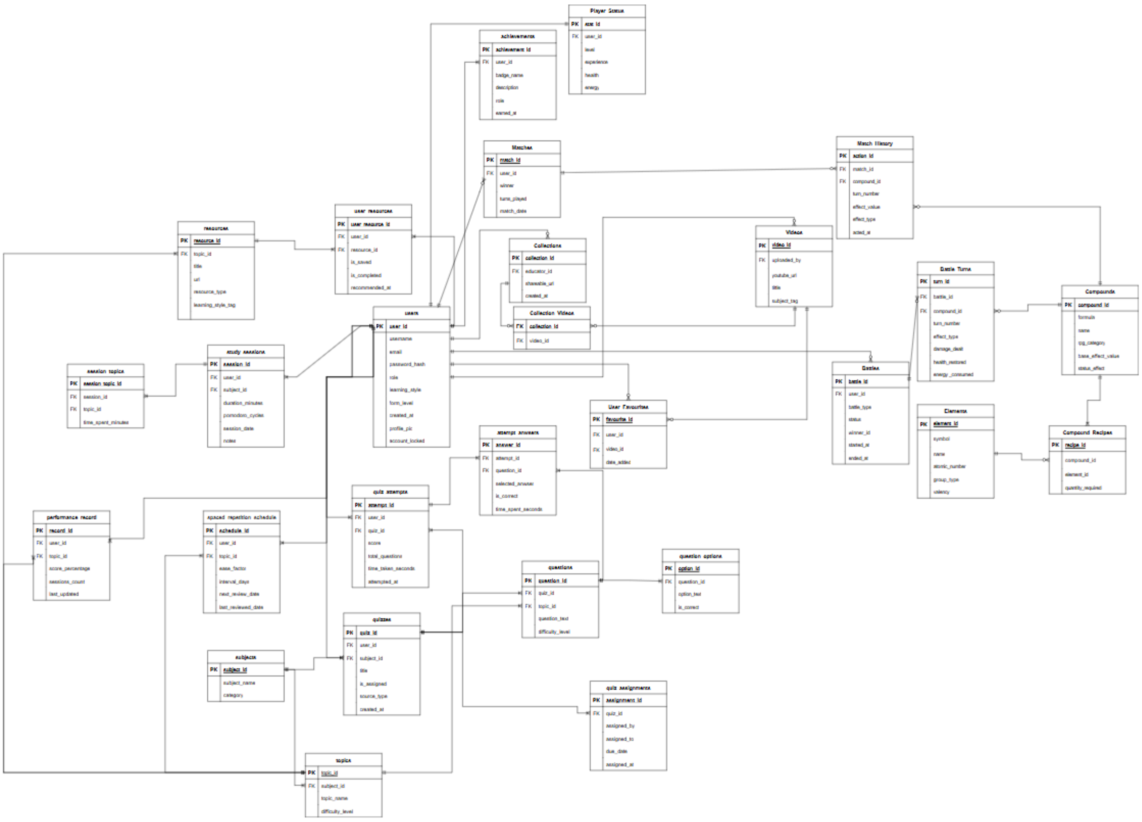


Figure 4.23: Entity Relationship Diagram of Qubo

4.5 Class Diagram

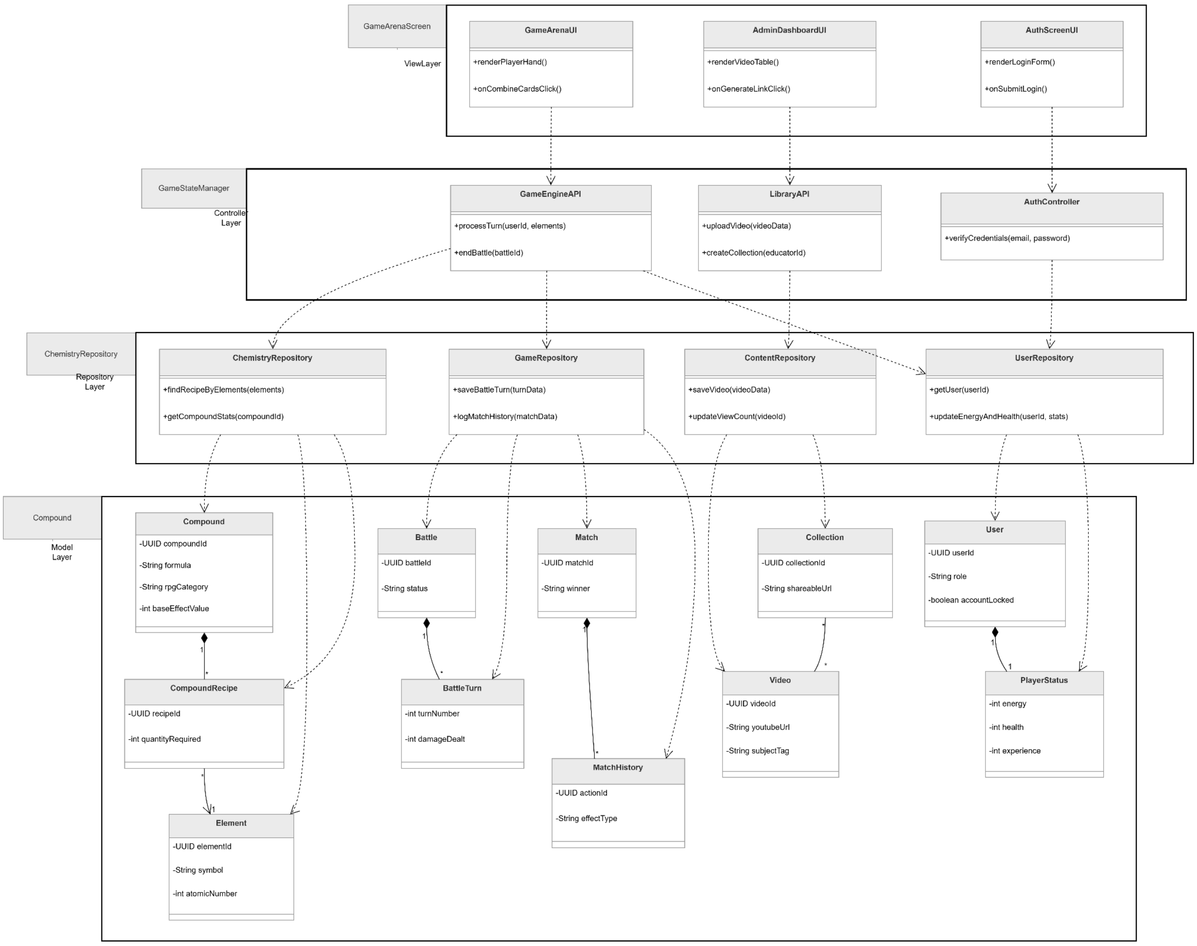


Figure 4.24: Class Diagram of Qubo Gamified interactive educational Learning Platform

4.6 Deployment Diagram

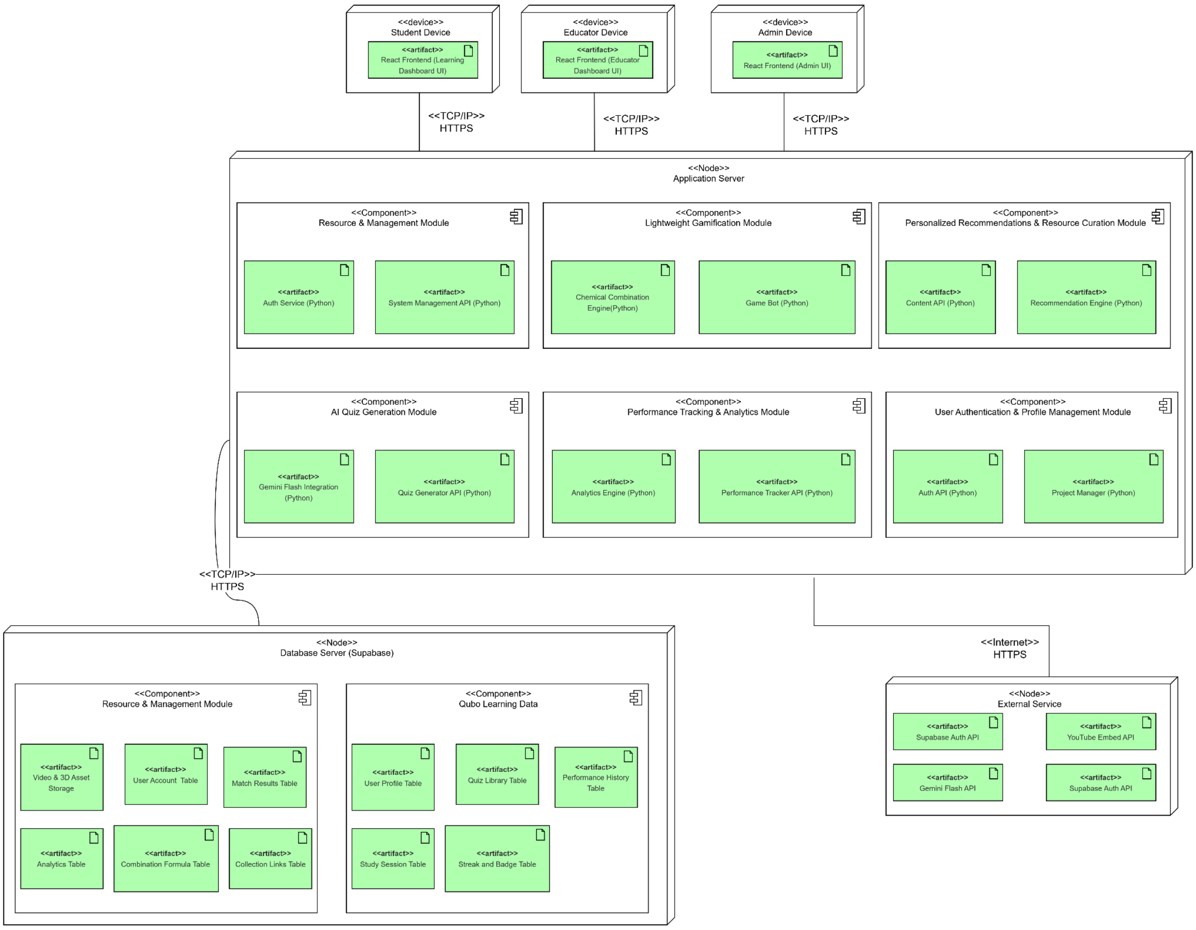


Figure 4.25: Deployment Diagram of Qubo

4.7 Chapter Summary and Evaluation

This chapter translated the core requirements from Chapter 3 into comprehensive technical blueprints for the Qubo platform. The User Interface design established modern, role-specific workflows, while Use Case and Sequence Diagrams mapped the behavioral logic between our React frontend, Python APIs, and Supabase database. Structurally, the ERD optimized complex data storage, the Layered Class Diagram mapped the application code, and the Deployment Diagram outlined a secure physical network architecture.

System Evaluation

Evaluating these blueprints, Qubo's architecture relies on the strict separation of concerns to ensure long-term maintainability. By utilizing a Layered Architecture, we decoupled the React UI components from Supabase database queries using Python API controllers, effectively shielding the frontend from future database modifications. This flexible structure directly supports scalability, particularly within the gamification engine. By abstracting core game mechanics into data-driven database tables rather than hardcoding them, we can introduce new content without triggering backend code deployments.

Alongside this scalability, the design enforces strict role boundaries and secure authentication protocols across the system to fulfill all critical security requirements. Ultimately, by translating complex backend operations into intuitive, role-specific UI dashboards, the architecture prevents information overload and delivers an optimized user experience. These blueprints provide a highly scalable, rock-solid foundation for the upcoming implementation phase.

Chapter 5

Implementation and Testing

5 Implementation and Testing

A short introduction that describes what will be included in this chapter.

IMPORTANT NOTE TO STUDENTS: Include details about:

- **Implementation**

  A detailed description of how you actually carried out the implementation (e.g., coding, etc.) of your system/prototype.

  - Include code snippets and descriptions to show how the requirements of the application/prototype have been met –
    - For Smart Campus projects, code snippets of the MQTT protocol for both client side and server side has to be included with explanation.
  - Include descriptions of important settings - Setting for server, network protocol, IP, database, security
  - Note: this should **not** be a chronological account of the work you carried out.

- **Testing**

  All test cases that have been carried out must be provided - To tabulate the test cases in tables

Note: Students may also opt to split Implementation and Testing into 2 separate chapters.

5.1 Sub-section 1 Heading

Sub-sections should be used to divide the chapter into logical parts.

5.1.1 Sub-subsection Heading

Sub-section numbering should be limited to a maximum of 3 levels (e.g. 5.3.1) in order to avoid confusion.

5.2 Sub-section 2 Heading

5.3 Sub-section 1 Heading

Sub-sections should be used to divide the chapter into logical parts.

5.3.1 Sub-subsection Heading

Sub-section numbering should be limited to a maximum of 3 levels (e.g. 5.3.1) in order to avoid confusion.

5.4 Sub-section 1 Heading

Sub-sections should be used to divide the chapter into logical parts.

5.4.1 Sub-subsection Heading

Sub-section numbering should be limited to a maximum of 3 levels (e.g. 5.3.1) in order to avoid confusion.

5.5 Chapter Summary and Evaluation

At the end of each chapter, evaluate the contents stated or discussed in the relevant sub-sections.

Chapter 6 *(if applicable)*

System Deployment

6 System Deployment

This chapter would be applicable for students who have embarked on a real-life industrial project. Students in this case would need to describe how the deployment has been carried out. Some of implementation tasks which need to be described include: training, file conversion or creation, and changeovers.

System Backup and Risk Management

Describe the procedures to backup the existing system for changeover purpose. Discuss the potential risk(s) for the changeovers and the solution.

On-site Setup

Describe the preparations to be done prior to the setup of the new system on client's site. Discuss the procedures to setup the new system, schedule, etc.

Training Procedure

Describe the procedures on the training procedures, contents, schedule, etc.

Follow-up

Describe the plan or procedures to follow up with the client (company) to verify the system reliability.

Chapter Summary and Evaluation

At the end of each chapter, evaluate the contents stated or discussed in the relevant sub-sections. For example,

- Problems faced. Describe the various problems faced by students in the course of doing the project.
- Solutions. What have been done to solve the problems?
- What tools and techniques have been used and reasons for using them.

Chapter 7

Discussions and Conclusion

7 Discussions and Conclusion

Each student is required to make an *evaluation of the project* he/she has embarked on. The project evaluation may include the following sections.

IMPORTANT NOTE TO STUDENTS: In this chapter, for problems related to code, hardware, internet connection, etc (where applicable):

- List the technical problems faced and state how they were resolved
- List the unsolved technical problems for future enhancement
- List the achieved objectives/modules
- List the incomplete parts for future enhancement - this is regardless the parts listed in the pre-determined scope. Suggestions for future improvement

Summary

Summarize the project including the problem and proposed solutions, justification of the choice of tools, techniques and methodologies used in this project.

Achievements

Students are required to evaluate the project's achievement against project objectives, completion of the project, students' view of the strengths and weaknesses of the work done.

Contributions

Discuss the creativity, innovativeness, contribution of the proposed system. Explain why the proposed system is necessary. Describe the marketability of the system.

Limitations and Future Improvements

Identify the limitations of the research or project. Provide suggestions for improvement or further development of the system or research in the future.

Issues and Solutions

Students are required to describe the various problems faced by students during the project development and explain what has been done to solve the problems. The valuable experiences gained or lessons learnt through the project as a whole. The issues may include technical issues, project management issues, team dynamics problems, and other difficulties encountered and lessons learnt. How the issues are solved or can be solved to ensure the project can be completed on time or to be improved in the future.

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Appendices

In order to enhance better understanding of the project, students should as far as possible include all directly relevant materials, figures or diagrams in the **main body** rather than in the Appendix. The appendix is reserved only for items which may not directly be relevant or essential to enhance a reader's understanding of the project, or which may interrupt the smooth reading of the project document (for example being too voluminous).

Appendices should only include supportive materials **directly referred** to in the writing and should be kept to a **minimum**, e.g. selected pages of an annual report, not the entire document. Examples of items included in Appendices are:

- Company's report and documentation, such as sample invoice, purchase order form, etc.
- Project meeting documentation e.g. minutes of meetings, tracking documents, memos etc.
- Questionnaires and results, interview questions and results, pilot test and results, observation sheet and results, experiment test plan and results, etc.
- Analysis/design diagrams (only those not incorporated in the main body of the report).

If there is more than one appendix, they should be identified as A, B, etc (e.g. Appendix A). Formulae and equations in appendices should be given separate numbering: Eq. (A.1), Eq. (A.2), etc.; in a subsequent appendix, Eq. (B.1) and so on. Similarly for tables and figures: Table A.1; Fig. A.1, etc.

IMPORTANT NOTE TO STUDENTS

APPENDIX *n* User Guide

- List the username and password of multiple roles (if applicable)
- Provide clear screen shots of each page, explain the functions of each button

As a rough guide, the user guide should include the following sections:

System Document

In this section, students should provide the following pieces of information:

- **System (hardware and software requirements)**. Students should describe the minimum hardware and software requirements to install the software application which has been developed by the students, for example, DBMS, OS, program development tools etc.
- **Installation**. Under this section, students should create an 'installation' CD and provide a brief step-by-step guide on how a new user can install the software on a computer system. Students should indicate any special setup information, such as the specific location of placing the database files, the SQL statements to add tables, etc.

Software source code should also be included in this CD. Students are not required to print out the software code.

Operation Document

Under this section, students are required to provide a brief step-by-step guide on how to use the installed software. The guides should teach the user how to run the software and use its major functions and features. For example, steps show guide the users on how to run the system, e.g. to use an executable file or to use the IDE. The login information such as username and password for each of the different users (or roles) must be provided. Some screen interfaces would be useful.

APPENDIX *n+1* Developer Guide*

*To be included for ***Smart Campus Projects*** and ***Real-Life Projects***. This section should include the following:

- List the necessary software, installer, API, library that must be installed
- List the authentication details, such as username and password for all security

Other Appendices:

- Supporting documents
- Non-disclosure agreement (if applicable)
- Other detailed documentation, such as important references, interview results, survey results, etc.

Book122.xlsx Gantt Chart

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